

Low rank is enough for the neural tangent kernel: depth and rank commutes towards stable training of deep networks

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Abstract

In the lazy regime, training deep neural networks reduces to kernel regression, and the spectrum of the neural tangent kernel (NTK) controls convergence and stability. Deep networks with low-rank bottlenecks (RF-LR) freeze random feature maps and train only narrow readouts of dimension $r \ll N$ per layer, achieving 95–99% fewer parameters than full-width MLPs at comparable depth. We study how depth L and bottleneck rank r shape this kernel in the sequential infinite-width limit and as $r \rightarrow \infty$. We derive an explicit NTK recursion for RF-LR with a visible $1/r$ factor at each bottleneck layer and a closed-form expansion at any depth. For the deterministic (proxy) kernel we prove sharp maximum eigenvalue linear in depth scaling and condition-number bounds: $\kappa \geq \Omega(r \cdot L)$ in general, and $\kappa_{\perp} = 1$ or $1 + o(1)$ for high-dimensional spherical data. Under a fixed parameter budget $O(NLr)$, depth and rank trade off and *commute* from a conditioning perspective: the maximum eigenvalue $\lambda_{\max}$ of the centered Gram matrix stays bounded, so the stable learning rate $2/\lambda_{\max}$ does not shrink with depth; one can thus choose deeper RF-LR with the same optimization bounds as shallower ones and retain the approximation benefits of depth. We show that *low rank does not shrink the kernel function class*: the three-layer mean RF-LR kernel induces the same RKHS as the shallow ReLU kernel (expressivity preserved, trainable parameters $O(LrN)$). We prove a rigorous proxy–empirical bound for equicorrelated data; for general data the concentration is sketched and left open. Numerical experiments confirm depth scaling, conditioning, and proxy–empirical concentration. All condition-number statements refer to the proxy kernel; RKHS equivalence holds for three layers, with $L \geq 4$ left to future work.

Keywords: Neural Tangent Kernel, Random Features, Low Rank, Depth Scaling, Conditioning

1. Introduction

In the lazy/NTK regime, training deep neural networks linearizes and reduces to kernel regression with an (approximately) fixed Gram matrix [Jacot et al. \(2018\)](#). To reduce parameters, one can factorize weights as $W = LR^{\top}$ with $r \ll \min\{n, m\}$; a central question is whether such factorizations preserve favorable optimization and convergence [Zhang et al. \(2025\)](#). We combine this with the random-feature lazy regime: we freeze the feature (left) directions and train only the readout (right) factors into a bottleneck of dimension r per layer. That combination defines the RF-LR architecture studied in this paper: RF-LR freezes random feature directions and trains only linear readouts (and biases) into a bottleneck of dimension $r \ll N$ at each layer. This paper asks: *how do depth L and bottleneck dimension r control conditioning-relevant scales for optimization under the NTK perspective?* We adopt the standard NTK setup (ReLU activation, isotropic Gaussian initialization, sequential infinite-width limit) and analyze the induced kernel at initialization.

The architecture stacks high-then-low width layers with a bottleneck $r \ll n_1$; we train readouts and biases while keeping feature directions frozen. In the kernel regime, the Gram matrix spectrum controls optimization rates [Jacot et al. \(2018\)](#); [Cao et al. \(2020\)](#). We characterize limiting

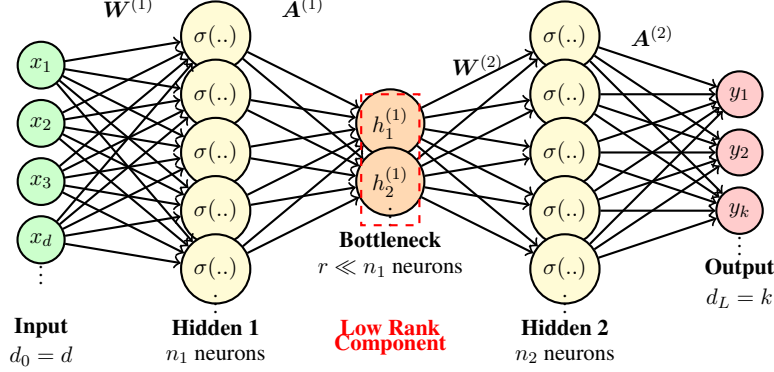


Figure 1: RF-LR: low-rank bottleneck (red dashed) reduces width from n_1 to r ; readouts trained, feature directions frozen.

kernels and spectra under a composition-of-GPs view in the sequential infinite-width limit. Figure 1 illustrates the layout.

The NTK framework [Jacot et al. \(2018\)](#) characterises infinitely wide networks as kernel methods. Extensions include finite-width corrections [Arora et al. \(2019\)](#); [Dyer and Gur-Ari \(2019\)](#), depth scaling [Yang \(2019\)](#); [Terjék \(2022\)](#), and, in the extensive-width regime $n \sim N^2$ with layers in the RMT limit, limits for Gram spectra [Benigni and Paquette \(2025\)](#). Low-rank architectures cut parameters and change optimisation dynamics [Sainath et al. \(2013\)](#); [Novikov et al. \(2015\)](#); [Arora et al. \(2018\)](#). Freezing feature directions isolates the bottleneck dimension r as the main control parameter. LoRA-style theory places low-rank fine-tuning between kernel and feature-learning regimes [Dayi and Chen \(2024\)](#); [Jang et al. \(2024\)](#). For ReLU kernels, Bietti and Bach [Bietti and Bach \(2021\)](#) showed that deep and shallow networks can induce the same RKHS via endpoint expansions. We combine Fisher–Kibble decoupling [Fisher \(1915\)](#); [Hotelling \(1953\)](#) with Puiseux analysis near $\rho = 1$ to get the same for the three-layer RF-LR mean kernel.

For MLPs at the edge of chaos, depth drives correlations toward 1; layerwise kernels become nearly constant and the deep mean kernel tends to rank-one [Terjék and González-Sánchez \(2025\)](#); [Hayou et al. \(2022\)](#). The NTK Gram matrix’s maximum eigenvalue scales linearly with depth, which can make training unstable. The same correlation alignment holds in RF-LR, but our proxy kernel has bounded $\lambda_{\max}$, so a fixed learning rate stays valid. The mean kernel’s infinite-depth limit still collapses after centering unless time or learning rate is rescaled (Appendix D.9). Freezing random features removes cross-layer dependencies present in fully trained nets; with Fisher–Kibble decoupling this yields explicit formulas and a path to joint large- n regimes in the RMT framework [Benigni and Paquette \(2025\)](#). Theorem 3 suggests organising finite-width corrections as an expansion in r . Future work may address tensor programs [Yang \(2020\)](#) and depth-dependent NTK spectra for MLPs at EOC [Terjék and González-Sánchez \(2025\)](#), with explicit bottleneck scaling.

In the lazy/NTK regime, gradient flow reduces to kernel regression with an (approximately) fixed kernel [Jacot et al. \(2018\)](#), and the spectrum of that kernel governs convergence and stability; we characterise the RF-LR NTK in the sequential infinite-width limit and establish the following contributions.

- **Explicit NTK recursion and closed form.** We derive the infinite-width NTK recursion for RF-LR with an explicit $1/r$ factor at each bottleneck layer and a closed-form L -layer expansion (Theorem 3, Corollary 4).
- **Depth scaling and condition number.** For the deterministic proxy kernel we prove sharp depth scaling: correlation alignment $1 - \rho_k = \Theta(k^{-2})$ (the same as for full-width MLPs at the edge of chaos Terjék and González-Sánchez (2025); Hayou et al. (2022)), kernel saturation to a fixed point, and diagonal–off-diagonal gap $\asymp 1/(rk)$ (Theorem 5). We prove a condition-number *lower bound* $\kappa \geq \Omega(r \cdot L)$ for the proxy Gram in general (Proposition 6), and *exact* conditioning $\kappa_{\perp} = 1$ or $\kappa_{\perp} = 1 + o(1)$ for equicorrelated or high-dimensional spherical data (Corollary 8). Under a fixed parameter budget $O(NLr)$, depth and rank thus trade off from a conditioning perspective.
- **RKHS equivalence (three layers).** We show that the mean three-layer RF-LR kernel induces the same RKHS as the shallow ReLU kernel (Corollary 13): the bottleneck does not shrink the kernel-regime function class, while reducing trainable parameters from $O(LN^2)$ to $O(LrN)$ per layer. The proof uses Fisher–Kibble decoupling and a Puiseux expansion at the endpoint; extension to $L \geq 4$ remains open.
- **Numerical experiments.** Appendix E presents deterministic and finite-width experiments that confirm the depth scaling (Theorem 5), equicorrelated and high-dimensional conditioning (Corollary 8), and proxy–empirical concentration (Theorem 7); non-equicorrelated data illustrate the $\kappa \geq \Omega(r \cdot L)$ lower bound (Proposition 6).

All condition-number bounds refer to the proxy Gram matrix; we give a rigorous proxy–empirical bound on $1^{\perp}$ for equicorrelated data (Theorem 7). Proof sketch: Appendix B.10.

Paper organization. Section 2 defines the RF-LR architecture, training policy, EOC parameterization, and the base and derivative kernels. Section 3 states the infinite-width NTK recursion and its closed-form L -layer expansion. Section 4 gives depth scaling for the deterministic proxy kernel and condition-number bounds (proxy and, for equicorrelated data, empirical). Section 5 proves that the three-layer mean RF-LR kernel induces the same RKHS as the shallow ReLU kernel via Fisher–Kibble decoupling and Puiseux analysis.

2. Network definition and EOC parameterization

We define the RF-LR architecture, the training policy, and the base and derivative kernels used in the NTK recursion.

2.1. RF-LR architecture

The layer-wise recurrence is as follows. Let $\mathbf{h}^{(0)}(x) = x \in \mathbb{R}^{d_0}$. For $\ell = 1, \dots, L$,

$$\mathbf{h}^{(\ell)}(x) = \frac{1}{\sqrt{n_{\ell}}} \sum_{j=1}^{n_{\ell}} A_j^{(\ell)} \sigma\left(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x)\right) + c^{(\ell)}. \quad (1)$$

Training policy: The readouts $A^{(\ell)}$ and layer biases $c^{(\ell)} \in \mathbb{R}^{d_{\ell}}$ are trained; the feature directions $w^{(\ell)}$ are frozen random draws (i.i.d. Gaussian). The $1/\sqrt{n_{\ell}}$ scaling (NTK parameterization) ensures a well-defined infinite-width limit.

Width vs bottleneck dimension. We distinguish the feature width n_ℓ (number of random features) from the output dimension $d_\ell = \dim(\mathbf{h}^{(\ell)})$. The trainable matrix $A^{(\ell)} \in \mathbb{R}^{d_\ell \times n_\ell}$ has columns $A_j^{(\ell)} \in \mathbb{R}^{d_\ell}$. A *bottleneck* layer corresponds to $d_\ell = r \ll n_\ell$, yielding $O(rn_\ell)$ trainable parameters and $\text{rank}(A^{(\ell)}) \leq r$ automatically.

Training parameters, biases, and initialization. Initialization is

$$w_j^{(\ell)} \sim \mathcal{N}(0, I_{d_{\ell-1}}/d_{\ell-1}), \quad A_{ij}^{(\ell)} \sim \mathcal{N}(0, 2), \quad c^{(\ell)} = 0.$$

We use the *edge-of-chaos (EOC)* scaling for the weights so that $\text{Cov}(w_j^{(\ell)}) = I_{d_{\ell-1}}/d_{\ell-1}$ Hayou et al. (2018). In our bottleneck regime we take $d_{\ell-1} = r$ for $\ell \geq 2$, so $w_j^{(\ell)} \in \mathbb{R}^r$ with covariance I_r/r .

We include a layer bias $c^{(\ell)}$ for completeness, but we avoid introducing extra scaling parameters for it. Bias scaling can be absorbed into the learning rate; the infinite-width and NTK structures remain qualitatively identical, and the kernel-regime predictions depend on kernel geometry, not absolute scales.

With this setup in place, we record the standing assumptions and conventions used throughout the paper.

2.2. Assumptions

The following conventions are in force throughout. We take the sequential infinite-width limit ($n_1 \rightarrow \infty, \dots, n_L \rightarrow \infty$) with i.i.d. Gaussian random features frozen after initialization under NTK scaling $1/\sqrt{n_\ell}$ Jacot et al. (2018); for $\ell \geq 2$ we assume the bottleneck regime $d_\ell = r \ll n_\ell$ with edge-of-chaos scaling $\text{Cov}(w_j^{(\ell)}) = I_r/r$. For RKHS and concentration we use normalized inputs (typically $x \in \mathbb{S}^{d-1}$) so kernels are zonal in the cosine similarity ρ , and we assume positive 1-homogeneity of the activation (satisfied by σ), which is used for Fisher–Kibble decoupling (Lemma 9) and radial/angular separation in the three-layer kernel. Full statements and discussion are in Appendix B (standing assumptions, EOC scaling, and Assumption 1).

2.3. Parameterization, signal propagation, and kernels

We briefly record the scaling conventions used in the kernel calculations; details are in Appendix B.4 and Appendix B.6.1. *EOC scaling*: we choose the variance σ_A so that the forward variance remains stable across depth (edge of chaos), leading to a recursion for the layerwise variance $q^{(\ell)}$ and a corresponding condition on σ_A . *Bottleneck scaling and the $1/r$ factor*: in the bottleneck regime $d_{\ell-1} = r$ for $\ell \geq 2$, the EOC scaling uses $\text{Cov}(w) = I_r/r$, which keeps base and derivative kernels $O(1)$, and the RF-LR NTK recursion carries an explicit prefactor $1/r$ at bottleneck layers. With an abuse of notation, we denote by $\Sigma^{(\ell)}$ the base kernel and by $\dot{\Sigma}^{(\ell)}$ the derivative kernel.

Definition 1 (Base and derivative kernels) *In the sequential infinite-width limit, the layer- ℓ base kernel is*

$$\Sigma^{(\ell)}(x, x') = \mathbb{E}_w \left[\sigma(w^\top \mathbf{h}^{(\ell-1)}(x)) \sigma(w^\top \mathbf{h}^{(\ell-1)}(x')) \right], \quad (2)$$

and the derivative kernel is

$$\dot{\Sigma}^{(\ell)}(x, x') = \mathbb{E}_w \left[\dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x)) \dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x')) \|w\|^2 \right]. \quad (3)$$

For $\ell = 1$ and centered Gaussian w , $\Sigma^{(1)}$ is rotation-invariant with standard closed forms in terms of input cosine similarity $\rho = \langle x, x' \rangle / (\|x\| \|x'\|)$:

$$\Sigma^{(1)}(x, x') = \frac{\|x\| \|x'\|}{2\pi} ((\pi - \theta) \cos \theta + \sin \theta), \quad \theta = \arccos(\rho).$$

We emphasize that these are not the NNGP kernel and the classical derivative kernel appearing in the NTK formulation. Rather, they are the base kernel and the derivative kernel of the Gaussian process $\mathbf{h}^{(\ell)}$ conditional on $\mathbf{h}^{(\ell-1)}$. By rotation invariance of Gaussian weights, these kernels depend on inputs only through the cosine similarity $\rho = \langle x, x' \rangle / (\|x\| \|x'\|)$. We will therefore write $\Sigma^{(\ell)}(\rho)$ and $\dot{\Sigma}^{(\ell)}(\rho)$. Given these definitions, we can now state the following composition result.

Theorem 2 (Gaussian process composition) *In the sequential infinite-width limit, the hidden states $\mathbf{h}^{(\ell)}$ form a composition of Gaussian processes:*

- *Conditionally on $\mathbf{h}^{(\ell-1)}$, for any finite collection of inputs $\{x_1, \dots, x_m\}$, the vector $(\mathbf{h}^{(\ell)}(x_1), \dots, \mathbf{h}^{(\ell)}(x_m))$ converges in distribution to a multivariate Gaussian.*
- *The conditional limiting Gaussian has mean zero and covariance matrix $[\Sigma^{(\ell)}(x_i, x_j)]_{i,j=1}^m$, where $\Sigma^{(\ell)}$ is the base kernel defined in Definition 1 and depends on $\mathbf{h}^{(\ell-1)}$.*

Proof See Appendix B.6.2. ■

3. NTK recursion

We now state the main result characterizing the NTK recursion for RF-LR. This recursion governs how the kernel evolves through layers and determines the training dynamics in the infinite-width limit.

Theorem 3 (Infinite-width NTK recursion) *Under NTK parameterization $1/\sqrt{n_\ell}$ and a sequential limit $n_1 \rightarrow \infty, \dots, n_L \rightarrow \infty$ with $\mathbf{w}^{(\ell)} \sim \mathcal{N}(0, \mathbf{I}_{\mathbf{d}_{\ell-1}}/\mathbf{d}_{\ell-1})$ i.i.d. frozen after initialization, the layer- ℓ NTK satisfies*

$$\Theta^{(0)}(x, x') = 0, \tag{4}$$

$$\Theta^{(1)}(x, x') = 1 + \Sigma^{(1)}(x, x'), \tag{5}$$

$$\Theta^{(\ell)}(x, x') = 1 + \frac{1}{r} \Theta^{(\ell-1)}(x, x') \cdot \dot{\Sigma}^{(\ell)}(x, x') + \frac{1}{r} \Sigma^{(\ell)}(x, x'), \quad \ell \geq 2, \tag{6}$$

where the factor $1/r$ appears for bottleneck layers $\ell \geq 2$ (low-rank regime). For $\ell = 1$, $\Sigma^{(1)}$ is deterministic; for $\ell > 1$, $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$ are random fields whose fluctuations shrink with the rank r of bottlenecks. The additive constant 1 term comes from the trained layer biases $c^{(\ell)}$ and corresponds to the constant mode. For mean-zero targets (or after centering labels), the relevant spectrum is that of the kernel restricted to $\mathbf{1}^\perp$, so $\kappa_\perp$ is the quantity governing convergence; the constant mode then plays no role.

Furthermore, for a finite-width network initialized at random parameters θ_0 , the empirical NTK $\Theta_{\theta_0}^{(L)}$ is a random kernel (a Gram operator of gradients). Consequently the kernel gradient flow trajectory $t \mapsto f_t$ is also a random object (measurable with respect to θ_0) and satisfies

$$\frac{d}{dt}f_t(x) = - \int \Theta_{\theta_0}^{(L)}(x, x')(f_t(x') - y(x'))d\mu(x'),$$

where μ is the data distribution and y is the target function. In the sequential infinite-width NTK limit, $\Theta_{\theta_0}^{(L)}$ concentrates and converges (in probability) to the deterministic kernel $\Theta^{(L)}$, yielding the corresponding deterministic limiting gradient flow.

Proof: See Appendix B.6.1, Proof B.6.1.

We can now derive an explicit closed-form expression for the NTK at any depth by expanding the recursion.

Corollary 4 (General L-layer NTK; closed form) *Under the assumptions of Theorem 3, for any $L \geq 1$, the NTK admits the explicit form:*

$$\Theta^{(L)}(x, x') = \sum_{\ell=1}^L \frac{1}{r^{L-\ell}} \left(\prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x') \right) + \sum_{\ell=1}^L \frac{1}{r^{\max(1, L-\ell)}} \Sigma^{(\ell)}(x, x') \prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x'), \quad (7)$$

where the first sum has bias contributions (empty product = 1 when $\ell = L$) and the second has fresh-basis contributions; each bottleneck layer contributes a factor $1/r$. For $L = 2$: $\Theta^{(2)} = 1 + \frac{1}{r}(\dot{\Sigma}^{(2)} + \Sigma^{(1)}\dot{\Sigma}^{(2)} + \Sigma^{(2)})$: the leading 1 is the bias (constant mode); $\frac{1}{r}\dot{\Sigma}^{(2)}$ is the bias-path term (derivative at layer 2 only); $\frac{1}{r}\Sigma^{(1)}\dot{\Sigma}^{(2)}$ and $\frac{1}{r}\Sigma^{(2)}$ are the fresh-basis terms (layers 1 and 2). This matches Theorem 3 by substituting $\Theta^{(1)} = 1 + \Sigma^{(1)}$ into the recursion for $\ell = 2$.

Proof Expanding the recursion in Theorem 3 yields the two sums in (7). A complete proof by induction (including an explicit $L = 3$ expansion) is given in Appendix B.6.4; see also Appendix 14. ■

Finite-width corrections and comparison with the full-width MLP NTK at EOC are given in Appendix B.5 and Appendix B.7 (Proposition 15).

3.1. Large-depth regime: probabilistic kernel and exponential decay

We summarize the probabilistic recursion and exponential depth suppression; full details are deferred to the appendix.

The mean (probabilistic) recursion and the resulting nested conditional expectations are essential for understanding the deep ($L \rightarrow \infty$) behavior, but the full discussion is lengthy. We therefore defer the probabilistic recursion, its nested-expectation form, and the exponential depth-suppression bound to Appendix B.8 (see also Appendix C.3 for related open problems). The three-layer case admits a compact representation that makes the Fisher–Kibble structure explicit; we present it in Section 5.1.

4. Depth scaling of the RF-LR NTK

We state sharp depth scaling for the deterministic proxy kernel and condition-number bounds; under a fixed budget $O(NLr)$, depth and rank commute from a conditioning perspective.

Random correlations and deterministic proxy. At each layer $\ell \geq 2$, the correlation $\rho_\ell = \cos \angle(\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x'))$ is a *random variable*: given the population correlation from the previous layer, Fisher’s law and Lemma 20 give $\mathbb{E}[(\rho_\ell - \varrho(\rho_{\ell-1}))^2] \leq C/r$ and $\mathbb{P}(|\rho_\ell - \varrho(\rho_{\ell-1})| \geq t) \leq 6 \exp(-c r t^2)$, so fluctuations are $O(1/\sqrt{r})$ sub-Gaussian. The NTK recursion passes through this chain of random correlations. The *deterministic proxy* replaces ρ_ℓ by the deterministic iterates $\rho_k = \varrho^{\circ(k-1)}(\rho_1)$ and analyzes the scalar recursion (84); for large r the random kernel concentrates around the proxy (Appendix B.8); a formal bound $\|\hat{K} - K_{\text{proxy}}\|_{\text{op}} = o_P(1)$ as $r \rightarrow \infty$ and its implication for empirical conditioning are given in Appendix C.4. In the equicorrelated case, a full proof yields $\|(\hat{K} - K_{\text{proxy}})|_{1^\perp}\|_{\text{op}} = O_P(L/r + 1/\sqrt{r})$ (Theorem 7). All condition-number statements in this section refer to the *proxy* Gram matrix; we give a lower bound on its condition number and, in one special case, exact conditioning with high probability.

Depth–rank scaling. The recursion contains an explicit $1/r$ bottleneck factor, which leads to saturation of the kernel magnitude as depth grows, while the diagonal–off-diagonal gap decays with depth in the proxy recursion (the same $1 - \rho_k = \Theta(k^{-2})$ correlation alignment as for full-width MLPs at the edge of chaos Terjék and González-Sánchez (2025); Hayou et al. (2022)). Theorem 5 analyzes this deterministic proxy, which approximates the mean path of the random recursion for large r . We refer to Appendix D.4 for notation and kernel expansions.

Theorem 5 (Depth scaling for the deterministic proxy RF-LR NTK) Fix $r > 1$ and $\rho_1 \in (-1, 1)$. Let $\Theta^{(k)}(\rho_1)$ be the deterministic proxy defined by (84), with $\rho_k = \varrho^{\circ(k-1)}(\rho_1)$ deterministic. Then (as for MLPs at EOC Terjék and González-Sánchez (2025); Hayou et al. (2022)):

- **(Correlation alignment)** The correlation recursion aligns polynomially:

$$1 - \rho_k = \Theta(k^{-2}).$$

- **(Kernel saturation)** $\Theta^{(k)}(\rho_1) \rightarrow \Theta_\star(r)$ as $k \rightarrow \infty$, where $\Theta_\star(r)$ is the fixed point of the limiting recursion obtained by setting $\dot{s} = s = 1/2$, namely

$$\Theta_\star(r) = \frac{2r + 1}{2r - 1}.$$

Moreover,

$$\Theta^{(k)}(\rho_1) - \Theta_\star(r) = O(k^{-1}).$$

- **(Depth-induced gap decay)** Let $\Theta_{\text{diag}}^{(k)} := \Theta^{(k)}(1)$ (on-diagonal value, i.e. identical inputs) and $\Theta_{\text{off}}^{(k)} := \Theta^{(k)}(\rho_1)$ (off-diagonal value for an input pair with initial cosine similarity ρ_1). Then

$$\Theta_{\text{diag}}^{(k)} - \Theta_{\text{off}}^{(k)} = \Theta\left(\frac{1}{r k}\right),$$

for large k .

Proof See Appendix D.7. ■

Proposition 6 (Lower bound on condition number for the proxy) *In the setting of Theorem 5, consider the deterministic proxy kernel $\Theta^{(L)}(\rho)$ and form the $n \times n$ Gram matrix $\mathbf{M}$ with entries $M_{ij} = \Theta^{(L)}(\rho_{ij})$ for pairwise input cosine similarities $\rho_{ij} = \langle x_i, x_j \rangle / (\|x_i\| \|x_j\|) \in (-1, 1)$. Let $\lambda_{\max}$ and $\lambda_{\min}$ denote the maximum and minimum eigenvalues of $\mathbf{M}$ restricted to $\mathbf{1}^\perp$. Assume: (i) $n \geq 3$; (ii) there exists a pair (i, j) with ρ_{ij} bounded away from 1; (iii) the dataset is not equicorrelated (the ρ_{ij} for $i \neq j$ are not all equal). Then $\lambda_{\max} = \Theta(1)$, $\lambda_{\min} \leq O(1/(rL))$, and the condition number $\kappa = \lambda_{\max}/\lambda_{\min}$ satisfies $\kappa \geq \Omega(r \cdot L)$. When assumption (iii) is violated (equicorrelated data), the proposition does not apply; instead Corollary 8 gives the best-case $\kappa_\perp = 1$ (and similarly for approximately equicorrelated high-dimensional data).*

Proof See Appendix D.5. ■

Theorem 7 (Operator norm on $\mathbf{1}^\perp$ in the equicorrelated case) *Fix $n, L \geq 1$, $r \geq 2$, and the equicorrelated setup $\rho_{1,ii} = 1$, $\rho_{1,ij} = \rho_0$ for $i \neq j$ with $\rho_0 \in (-1, 1)$. Let $\hat{K}$ be the $n \times n$ empirical RF-LR NTK Gram matrix and K_{proxy} the deterministic proxy with entries $(K_{\text{proxy}})_{ij} = \Theta^{(L)}(\rho_{1,ij})$ (Definition D.4). Then there exist constants $C, c > 0$ (depending on L, ρ_0) such that for any $\epsilon > 0$,*

$$\mathbb{P}\left(\|(\hat{K} - K_{\text{proxy}})|_{\mathbf{1}^\perp}\|_{\text{op}} \geq \epsilon\right) \leq C \exp\left(-c \frac{r \epsilon^2}{L^2}\right) + C \exp\left(-c \frac{r \epsilon}{L}\right).$$

In particular, $\|(\hat{K} - K_{\text{proxy}})|_{\mathbf{1}^\perp}\|_{\text{op}} = O_P(L/r + 1/\sqrt{r})$ as $r \rightarrow \infty$ with n, L fixed. Hence if $r = O(L^2)$, then $\|(\hat{K} - K_{\text{proxy}})|_{\mathbf{1}^\perp}\|_{\text{op}} = O_P(L/r)$.

Proof See Appendix C.4. ■

Corollary 8 (Exact conditioning for equicorrelated and high-dimensional random data) *In the setting of Proposition 6, let $\kappa_\perp = \lambda_{\max}(\mathbf{M}|_{\mathbf{1}^\perp})/\lambda_{\min}(\mathbf{M}|_{\mathbf{1}^\perp})$ denote the condition number on the mean-zero subspace. The following hold.*

- **Equicorrelated data.** *If $\rho_{ij} = \rho_0$ for all $i \neq j$ (and $\rho_{ii} = 1$), then on $\mathbf{1}^\perp$ all $n - 1$ eigenvalues of $\mathbf{M}$ equal $\lambda_\perp = \Theta^{(L)}(1) - \Theta^{(L)}(\rho_0) = \Theta(1/(rL))$. Hence*

$$\kappa_\perp = \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{\lambda_\perp}{\lambda_\perp} = 1.$$

- **High-dimensional random data.** *If the inputs $x_1, \dots, x_n \in \mathbb{S}^{d-1}$ are drawn i.i.d. uniformly from the unit sphere, then as $d \rightarrow \infty$ (with n, r, L fixed) with high probability $\max_{i \neq j} |\rho_{ij}| = O(1/\sqrt{d})$, so the proxy Gram matrix is approximately equicorrelated with $\rho_0 = O(1/\sqrt{d})$. On that event, all eigenvalues on $\mathbf{1}^\perp$ are $\lambda_\perp = \Theta^{(L)}(1) - \Theta^{(L)}(\bar{\rho}) + o(1) = \Theta(1/(rL))(1 + o(1))$ with $\bar{\rho} = O(1/\sqrt{d})$, and they differ from each other by $o(\lambda_\perp)$. Hence*

$$\kappa_\perp = \frac{\lambda_{\max}}{\lambda_{\min}} = 1 + o(1) \quad \text{as } d \rightarrow \infty.$$

On that high-probability event, the empirical Gram matrix $\hat{K}$ concentrates around the proxy: $\|(\hat{K} - K_{\text{proxy}})|_{\mathbf{1}^\perp}\|_{\text{op}} = O_P(L/r + 1/\sqrt{r})$ (Theorem 7 with $\rho_0 = O(1/\sqrt{d})$).

Proof See Appendix D.6. ■

Limited scope of exact conditioning. Equicorrelated and high-dimensional i.i.d. spherical data are special; the general conditioning guarantee in this section is the proxy lower bound (Proposition 6). Summary of condition-number scaling:

- **Equicorrelated:** $\kappa_{\perp} = 1$ (exact; all eigenvalues on $\mathbf{1}^{\perp}$ equal $\lambda_{\perp} = \Theta(1/(rL))$).
- **High-dim i.i.d. spherical:** $\kappa_{\perp} = 1 + o(1)$ as $d \rightarrow \infty$ (approximately equicorrelated).
- **General (non-equicorrelated):** $\kappa \geq \Omega(r \cdot L)$; explicit: $\lambda_{\max} = \Theta(1)$, $\lambda_{\min} \leq O(1/(rL))$, hence $\kappa = \lambda_{\max}/\lambda_{\min} \geq \Omega(r \cdot L)$.

Proxy vs. optimization. Proposition 6 gives a *lower bound* on the condition number of the proxy Gram matrix; we see strong experimental agreement with our theory and proxy predictions, and exact RKHS equivalence holds for three layers, with extension to depth $L \geq 4$ left to future work. For the proxy and non-equicorrelated data, depth drives the kernel toward near-constant entries (gap $\Theta(1/(rL))$, as for MLPs at EOC) and $\kappa \geq \Omega(rL)$. The actual network uses random correlations with $O(1/\sqrt{r})$ sub-Gaussian fluctuations (Appendix B.8), so for large r the random kernel concentrates around the proxy. The condition number is relevant for kernel regression (stability and convergence speed depend on $\lambda_{\max}/\lambda_{\min}$); under a fixed parameter budget $O(NLr)$, depth and rank trade off from a conditioning perspective, and the choice of r and L is a matter of approximation power (effective depth and RKHS expressivity).

5. Low rank is enough for the NTK RKHS (three-layer mean kernel)

We show that low rank does not shrink the kernel-regime function class: the mean three-layer RFLR kernel induces the same RKHS as the shallow ReLU kernel.

Roadmap. We first derive the Fisher–Kibble decoupling (Section 5.1), which separates angular (correlation) and radial (norm) randomness and yields a compact three-layer empirical NTK form. We then obtain a Puiseux expansion of the mean kernel near $\rho = 1$ (Section 5.2), which controls the RKHS. The leading exponent matches the shallow ReLU kernel, so the RKHSs coincide (Corollary 13). The analysis is restricted to three layers because the Fisher–Kibble integral admits a closed form for a single bottleneck; for $L \geq 4$ (depth at least four layers), the nested Fisher-chain expectations require a Laplace-type expansion that remains open (Appendix C.3).

5.1. Microscopic distributions: Fisher–Kibble decoupling

We can now state a compact representation under a homogeneity assumption.

Lemma 9 (Fisher–Kibble decoupling) *Let x, y be input vectors and let x_1, y_1 denote their rank- r random projections. Define the empirical correlation $\rho_1 = \cos \angle(x_1, y_1)$ and squared norms $u = \|x_1\|^2, v = \|y_1\|^2$. Then:*

- ρ_1 follows Fisher’s correlation distribution [Fisher \(1915\)](#); [Hotelling \(1953\)](#) (centered at ρ). For $r > 2$:

$$p_{\text{Fisher}}(\rho_1) = \frac{(r-2) \Gamma(r-1) (1-\rho^2)^{\frac{r-1}{2}} (1-\rho_1^2)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r-\frac{1}{2}) (1-\rho\rho_1)^{r-\frac{3}{2}}} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r-\frac{1}{2}; \frac{1+\rho\rho_1}{2}\right).$$

- (u, v) follow Kibble's [Kibble \(1941\)](#) bivariate chi-square law ($u, v \geq 0$; I_ν is the modified Bessel function):

$$p_{\text{Kibble}}(u, v) = \frac{(uv)^{\frac{r}{4}-\frac{1}{2}} \exp\left(-\frac{u+v}{2(1-\rho^2)}\right)}{\Gamma(r/2) (2(1-\rho^2))^{\frac{r}{2}+1} \rho^{\frac{r}{4}-\frac{1}{2}}} I_{\frac{r}{2}-1}\left(\frac{\rho\sqrt{uv}}{1-\rho^2}\right).$$

- *Angular and radial parts are independent:* $p(\rho_1, u, v) = p_{\text{Fisher}}(\rho_1) p_{\text{Kibble}}(u, v)$.

Proof: See Appendix [C.2](#), Proof [C.2](#).

Homogeneity assumption enabling decoupling. By positive 1-homogeneity and rotational invariance, the base kernel factorizes and $(\rho_1, \|x_1\|, \|y_1\|)$ decouple into angular (Fisher) and radial (Kibble) components; see Appendix [19](#) and Appendix [C.2](#) for details, explicit densities, and the induced compact three-layer empirical NTK form.

Compact three-layer empirical NTK form. Under the assumptions of Theorem [3](#) with $L = 2$, the three-layer empirical RF-LR NTK admits the representation

$$\Theta^{(2)}(x, x') = 1 + \frac{1}{r} \Theta^{(1)}(x, x') \dot{\Sigma}^{(2)}(x, x') + \frac{1}{r} \Sigma^{(2)}(x, x'), \quad \Theta^{(1)}(x, x') = 1 + \Sigma^{(1)}(x, x'), \quad (8)$$

and, under the positive 1-homogeneity assumption (satisfied by σ), the layer-2 fields can be written in terms of the Fisher angular variable ρ_1 and the Kibble radial factor $w_r := \sqrt{uv}/r$ as

$$\dot{\Sigma}^{(2)}(x, x') = 1 - \frac{\arccos(\rho_1)}{\pi}, \quad \Sigma^{(2)}(x, x') = w_r \Sigma^{(1)}(\rho_1). \quad (9)$$

Combining (8)–(9) yields an explicit three-layer kernel in the decoupled microscopic variables (ρ_1, u, v) ; equivalently,

$$\Theta^{(2)}(x, x') = 1 + \frac{1}{r} \Theta^{(1)}(x, x') \left(1 - \frac{\arccos(\rho_1)}{\pi}\right) + \frac{1}{r} w_r \Sigma^{(1)}(\rho_1). \quad (10)$$

See Remark [19](#) (Appendix [C.2](#)) for the full derivation, the explicit Fisher and Kibble densities, and the induced compact form (10).

5.2. Endpoint expansion via hypergeometric analysis

The RKHS of zonal kernels on the sphere is controlled by the Puiseux behavior near $\rho = \pm 1$ [Bietti and Bach \(2021\)](#). For the mean three-layer kernel, the key step is to analyze the Fisher expectation $\mathbb{E}[\arccos(\hat{\rho}_r)]$ as $\rho \rightarrow 1$.

Proposition 10 (Puiseux expansion of $\mathbb{E}(\arccos(\hat{\rho}_r))$ near $\rho = 1$) *Let $r > 2$ and $\hat{\rho}_r \sim \text{Fisher}(\rho, r)$. Write $\rho = 1 - t$ with $t \downarrow 0$. Then*

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \sqrt{2t} I(r) + O(t^{3/2}),$$

where $I(r) \in (0, \infty)$ admits the closed form

$$I(r) = \frac{(r-2) 2^{r-\frac{5}{2}}}{\sqrt{2\pi}} \frac{\Gamma(\frac{r-1}{2}) \Gamma(\frac{r}{2}-1)}{\Gamma(r-1)} C_1(r), \quad C_1(r) = \frac{\Gamma(r-\frac{1}{2}) \Gamma(r-\frac{3}{2})}{\Gamma(r-1)^2}. \quad (11)$$

Proof [] Insert Fisher’s density (Appendix 19) into $\int_{-1}^1 \arccos(s) p_{\text{Fisher}}(s \mid 1-t, r) ds$ and change variables $s = 1 - v$. Near $t, v \rightarrow 0$, use $\arccos(1 - v) = \sqrt{2v} + O(v^{3/2})$ and apply the hypergeometric connection formula (DLMF §15.8.2 Olver et al. (2024)) to replace ${}_2F_1(\frac{1}{2}, \frac{1}{2}; r - \frac{1}{2}; 1 - \frac{t+v}{2})$ by the constant $C_1(r)$ up to $O((t+v)^{r-3/2})$. Then the leading integral reduces, after $w = v/t$, to a Beta integral and yields (11). See Appendix C.7 for the full proof. ■

Proposition 11 (*$I(r)$ decays as $1/\sqrt{r}$*) Let $I(r)$ be defined by (11). Then $I(r) = O(1/\sqrt{r})$ as $r \rightarrow \infty$.

Proof [Proof sketch] Apply Stirling’s formula to the Gamma ratios in (11). A detailed proof is given in Proposition 30 (Appendix C.9). ■

Corollary 12 (Puisseux expansion of the mean three-layer NTK near $\rho = 1$) Let $\tilde{\Theta}^{(2)}(\rho)$ denote the mean three-layer RF-LR NTK and let $K_\infty(\rho)$ denote the deterministic $r \rightarrow \infty$ limit kernel,

$$K_\infty(\rho) = \Theta^{(1)}(\rho) \left(1 - \frac{\arccos(\rho)}{\pi} \right) + \Sigma^{(1)}(\rho) + 1,$$

as in Appendix C.7. Writing $\rho = 1 - t$ with $t \downarrow 0$, one has

$$\tilde{\Theta}^{(2)}(1-t) = 1 + \frac{1}{r} (K_\infty(1) - 1) - \frac{1}{r} \left[\frac{2}{\pi} + \frac{\sqrt{2}}{2\pi} I(r) \right] t^{1/2} + O(t^{3/2}), \quad (12)$$

where $I(r)$ is as in Proposition 10. In particular, since $I(r) = O(1/\sqrt{r})$ (Proposition 11), the bracket equals $2/\pi + O(1/\sqrt{r})$, so the $t^{1/2}$ coefficient is $\frac{2}{\pi r} + O(r^{-3/2})$.

Proof [Proof sketch] Combine the three-layer mean-kernel identity (Appendix C.7) with Proposition 10 and the ReLU endpoint expansion $\arccos(1-t) = \sqrt{2t} + O(t^{3/2})$. A complete proof is given in Appendix C.7. ■

Interpretation: $1/r$ vs. $I(r)$. The $1/r$ prefactor in (12) comes from the EOC initialization and the RF-LR recursion: each bottleneck layer contributes $1/r$ (Section 2, Appendix B.4). With unnormalized classical NTK (no bottleneck scaling), the Puisseux coefficient would be $[\frac{2}{\pi} + \frac{\sqrt{2}}{2\pi} I(r)]$ and the $I(r) \sim 1/\sqrt{r}$ term would directly reflect the Fisher–Kibble randomness. Thus the $1/r$ decay in the expansion is due to EOC; $I(r) \sim 1/\sqrt{r}$ is the signature of the randomness-induced deviation from the deterministic limit.

5.3. RKHS equivalence via endpoint behavior

We state the main results and the RKHS identification.

Main results. For zonal kernels on the sphere, the RKHS is determined by the Puiseux exponents at the endpoints $\rho = \pm 1$ [Bietti and Bach \(2021\)](#). Our main results in this section are:

- **Puiseux expansion:** The mean three-layer NTK $\tilde{\Theta}^{(2)}(\rho)$ has a leading $t^{1/2}$ term near $\rho = 1 - t$ (Corollary 12). The coefficient involves $\mathbb{E}[\arccos(\hat{\rho}_r)]$, whose expansion uses the hypergeometric connection formula (DLMF §15.8.2 [Olver et al. \(2024\)](#)) applied to the ${}_2F_1$ in Fisher’s density (Proposition 10).
- **$I(r)$ scaling:** $I(r) = O(1/\sqrt{r})$ as $r \rightarrow \infty$ (Proposition 11), so the $t^{1/2}$ coefficient is an $O(1)$ perturbation of the shallow ReLU constant.
- **RKHS equivalence:** The three-layer mean kernel induces the same RKHS as the shallow ReLU kernel (Corollary 13 below), by the Bietti–Bach criterion ([Bietti and Bach, 2021](#), Theorem 1): matching the exponent $1/2$ at $\rho = 1$ suffices.

See Appendix C.7 for the full proof of the Puiseux expansion and RKHS identification.

Corollary 13 (Low rank is enough for the NTK RKHS: three-layer mean kernel) *Under the RF-LR setting with ReLU nonlinearity and isotropic random features on $\mathbb{S}^{d-1}$, the three-layer mean NTK $\tilde{\Theta}^{(2)}$ induces the same RKHS as the shallow ReLU kernel. In particular, the corresponding RKHSs coincide as sets with equivalent norms.*

Proof [Proof sketch] The mean three-layer kernel is zonal. Proposition 10 gives a Puiseux expansion near $\rho = 1$ with leading exponent $t^{1/2}$; Proposition 11 shows the r -dependent coefficient is an $O(1)$ perturbation of the shallow ReLU coefficient. Since the shallow ReLU kernel has the same endpoint exponent $1/2$, the Bietti–Bach criterion [Bietti and Bach \(2021\)](#) yields RKHS equivalence. See Appendix C.7 for a detailed argument. ■

6. Conclusion and discussion

A main takeaway is that deeper networks are not necessarily unstable in the lazy regime: in the low-rank kernel setting (RF-LR), bounded $\lambda_{\max}$ preserves step-size stability, and one can increase depth without sacrificing optimisation stability while retaining the benefits of deeper approximations that one can seize. We showed that depth and rank *commute* from a conditioning perspective—the explicit NTK recursion, sharp depth scaling, and condition-number bounds for the proxy kernel support this—and that low rank does not shrink the kernel function class (RKHS equivalence for three layers). The parameter budget $O(NLr)$ thus allows trading width for depth without losing expressivity or stable step sizes, which we validated numerically for depth scaling, conditioning, and proxy–empirical concentration on equicorrelated and spherical data. In short, RF-LR offers a way to reap the benefits of depth in the NTK regime at a fraction of the parameters of full-width MLPs. We hope that this work will encourage practitioners and theorists alike to adopt low-rank structures more widely, where the parameter–depth tradeoff (with 95–99% fewer parameters) and stability guarantees make them attractive. Proving that the same RKHS is induced for $L \geq 4$ layers is left for future work, as are a full proxy–empirical concentration proof for arbitrary data, the connection to gradient-descent convergence and sample complexity, bulk spectral laws in the RMT scaling, and finite-width corrections.

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Appendix A. Summary of main results

This work develops an NTK analysis for low-rank random-feature architectures (RF-LR) under explicit assumptions. Appendix E reports numerical illustrations that accompany the theory.

Main: depth/rank scaling and condition number. The trainable parameter budget scales as $O(NLr)$. In the deterministic proxy mean recursion, depth drives correlations toward 1 with $1 - \rho_k = \Theta(k^{-2})$ (the same correlation alignment as for full-width MLPs at the edge of chaos [Terjék and González-Sánchez \(2025\)](#); [Hayou et al. \(2022\)](#)), the kernel magnitude saturates, and the diagonal–off-diagonal gap decays as $\asymp 1/(rk)$ (Theorem 5). We prove a lower bound on the condition number of the *proxy* Gram matrix, $\kappa \geq \Omega(r \cdot L)$ (Proposition 6), and exact conditioning $\kappa_{\perp} = 1$ or $\kappa_{\perp} = 1 + o(1)$ for equicorrelated or high-dimensional random data (Corollary 8). Under a fixed budget $O(NLr)$, depth and rank trade off from a conditioning perspective; the choice of r and L is a matter of approximation power (e.g. effective depth and RKHS expressivity).

Recursion. We provide an explicit NTK recursion for arbitrary depth in RF-LR and a closed-form expansion (Theorem 3, Corollary 4).

Low rank is enough for the NTK RKHS (three layers). In the NTK regime, the three-layer RF-LR mean kernel induces the same RKHS as the shallow ReLU kernel (Corollary 13), showing that the bottleneck does not reduce the kernel-regime function class while lowering the trainable parameter scaling from dense $O(LN^2)$ to $O(LrN)$.

Concentration in the bottleneck dimension. For the three-layer case, the empirical kernel concentrates around its deterministic limit $K_{\infty}(\rho)$ with tails controlled by the bottleneck dimension r (Theorem 25, Corollary 26). For equicorrelated data, a full proof in the appendix gives $\|(\hat{K} - K_{\text{proxy}})|_{1^{\perp}}\|_{\text{op}} = O_P(L/r + 1/\sqrt{r})$ (Theorem 7), linking proxy and empirical conditioning on the mean-zero subspace.

Appendix B. Notation and standing assumptions

B.1. Notation

We collect all notation used in the main text and appendix.

Architecture and dimensions. L : depth (number of layers). ℓ : layer index, $\ell = 0, 1, \dots, L$. $\mathbf{n}_\ell$: width (number of random features) at layer ℓ . $\mathbf{d}_\ell$: output dimension at layer ℓ , with $\mathbf{d}_0 = d$ (input dimension) and $\mathbf{d}_\ell = r$ for $\ell \geq 1$ in the bottleneck regime. r : bottleneck rank, $r \ll \mathbf{n}_\ell$ for $\ell \geq 2$. N : generic notation for width (e.g. n_1 or $\mathbf{n}_\ell$). $\mathbb{S}^{d-1}$: unit sphere in $\mathbb{R}^d$.

Parameters and activations. $A^{(\ell)} \in \mathbb{R}^{\mathbf{d}_\ell \times \mathbf{n}_\ell}$: trainable readout matrix at layer ℓ ; columns $A_j^{(\ell)}$. $w_j^{(\ell)} \in \mathbb{R}^{\mathbf{d}_{\ell-1}}$: frozen random feature direction at layer ℓ , neuron j . $c^{(\ell)} \in \mathbb{R}^{\mathbf{d}_\ell}$: trainable layer bias. σ, σ' : activation function (ReLU). σ_A : readout variance scale; at EOC, $\sigma_A = \sqrt{2}$. $\Sigma_w = \text{Cov}(w_j^{(\ell)})$: covariance of feature weights; at EOC, $\Sigma_w = I_{\mathbf{d}_{\ell-1}}/\mathbf{d}_{\ell-1}$, and $\Sigma_w = I_r/r$ in the bottleneck regime.

Hidden states and kernels. $\mathbf{h}^{(\ell)}(x)$: hidden state at layer ℓ for input x . $\Sigma^{(\ell)}(x, x')$: base kernel (conditional covariance of $\mathbf{h}^{(\ell)}$ given $\mathbf{h}^{(\ell-1)}$). $\dot{\Sigma}^{(\ell)}(x, x')$: derivative kernel (conditional expectation involving $\dot{\sigma}$ and $\|w\|^2$). $\Theta^{(\ell)}(x, x')$ or $\bar{\Theta}^{(\ell)}(x, x')$: neural tangent kernel at layer ℓ (infinite-width limit). $\Theta_{\theta_0}^{(L)}$: empirical NTK for finite-width network with initial parameters θ_0 . $\bar{\Theta}^{(\ell)}$: mean (over randomness) NTK. $s(\rho), \dot{s}(\rho)$: scalar ReLU base and derivative kernels as functions of cosine similarity ρ . $\bar{\Sigma}^{(\ell)}(\rho)$: normalized derivative kernel as function of ρ . $K_\infty(\rho)$: deterministic $r \rightarrow \infty$ limit kernel (three-layer case). $q^{(\ell)} = \mathbb{E}\|\mathbf{h}^{(\ell)}(x)\|^2$: forward variance at layer ℓ .

Correlations and proxy. $\rho = \langle x, x' \rangle / (\|x\| \|x'\|)$: input cosine similarity. ρ_ℓ : (random) layer- ℓ correlation $\cos \angle(\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x'))$. $\rho_k = \varrho^{(k-1)}(\rho_1)$: deterministic proxy correlation at depth k . ϱ : ReLU cosine map (EOC correlation propagation). $\theta = \arccos(\rho)$: angle. $\Theta^{(k)}(\rho_1)$: deterministic proxy NTK at depth k . $\Theta_\star(r)$: fixed point of proxy recursion as $k \rightarrow \infty$. K_{proxy} : proxy Gram matrix with entries $\Theta^{(L)}(\rho_{ij})$. $\tilde{K}$: empirical RF-LR NTK Gram matrix.

Data and optimization. μ : data distribution on the input space (or on input–output pairs). $y(x)$: target function. n : number of training samples when considering a dataset $\{(x_i, y_i)\}_{i=1}^n$. $f_t(x)$: kernel gradient flow output at time t . θ_0 : initial parameters.

Gram matrix and spectrum. $\mathbf{M}$: Gram matrix with entries $M_{ij} = \Theta^{(L)}(\rho_{ij})$ for pairwise input similarities ρ_{ij} . $\lambda_{\max}, \lambda_{\min}$: maximum and minimum eigenvalues of $\mathbf{M}$ (or of its restriction to $\mathbf{1}^\perp$). $\kappa = \lambda_{\max}/\lambda_{\min}$: condition number. $\kappa_\perp$: condition number of $\mathbf{M}$ restricted to the mean-zero subspace $\mathbf{1}^\perp$. $\mathbf{1}$: all-ones vector; $\mathbf{1}^\perp$: subspace orthogonal to $\mathbf{1}$ (mean-zero subspace).

RKHS and Fisher–Kibble. $\mathcal{H}_\Theta, \mathcal{H}_{\tilde{K}}$: RKHS induced by kernel Θ or $\tilde{K}$. $I(r)$: coefficient in Puiseux expansion of $\mathbb{E}[\arccos(\hat{\rho}_r)]$ near $\rho = 1$. $\hat{\rho}_r$: sample correlation (Fisher distribution). $w_r = \sqrt{uv}/r$: Kibble radial factor (with $u = \|x_1\|^2, v = \|y_1\|^2$ for rank- r projections). $p_{\text{Fisher}}, p_{\text{Kibble}}$: Fisher and Kibble densities.

Asymptotics and constants. $O(\cdot), \Theta(\cdot), \Omega(\cdot)$: standard asymptotic notation. $o_P(1), O_P(\cdot)$: in probability. C, c, C_0, c_0 , etc.: positive constants that may depend on L, ρ_0 or other fixed parameters.

B.2. Standing assumptions (expanded)

We summarize the conventions used throughout the main text.

- **Sequential infinite width.** Widths $n_1, \dots, n_L$ tend to infinity sequentially under NTK scaling $1/\sqrt{n_\ell}$. Random features $w^{(\ell)}$ are i.i.d. Gaussian and frozen after initialization.
- **Training policy.** Only the readouts $A^{(\ell)}$ (and layer biases $c^{(\ell)}$) are trained; all other parameters are frozen.
- **Bottleneck regime.** For $\ell \geq 2$, the representation dimension is $d_\ell = r \ll n_\ell$. This induces an explicit $1/r$ factor in the RF-LR NTK recursion at bottleneck layers.
- **Zonal kernels.** By rotational invariance of Gaussian features, base and derivative kernels depend on inputs only through the cosine similarity $\rho = \langle x, x' \rangle / (\|x\| \|x'\|)$. For RKHS results we typically take $x \in \mathbb{S}^{d-1}$.
- **Homogeneity.** We use Assumption 1 (positive 1-homogeneity) for Fisher–Kibble decoupling and for separating radial and angular components in three-layer calculations.
- **Training data.** We assume training data are drawn from a distribution μ on the input space (or on input–output pairs). For kernel gradient flow and condition-number statements we assume sufficiently many training samples so that the empirical Gram matrix and its spectrum approximate the population (kernel) quantities; precise sample-complexity bounds are left to future work.

B.3. Finite-width empirical NTK and limiting gradient flow

For a finite-width network initialized at random parameters θ_0 , the empirical NTK $\Theta_{\theta_0}^{(L)}$ is a random kernel (the Gram operator of gradients $\langle \nabla_\theta f(x; \theta_0), \nabla_\theta f(x'; \theta_0) \rangle$). The kernel gradient flow trajectory $t \mapsto f_t$ is therefore a random object (measurable with respect to θ_0) and satisfies

$$\frac{d}{dt} f_t(x) = - \int \Theta_{\theta_0}^{(L)}(x, x') (f_t(x') - y(x')) d\mu(x'),$$

where μ is the data distribution and y is the target function. In the sequential infinite-width NTK limit, $\Theta_{\theta_0}^{(L)}$ concentrates and converges in probability to the deterministic kernel $\Theta^{(L)}$, yielding the corresponding deterministic limiting gradient flow driven by $\Theta^{(L)}$ [Jacot et al. \(2018\)](#).

Assumption 1 (Homogeneous activation) *The activation function σ is positively 1-homogeneous:*

$$\sigma(\alpha u) = \alpha \sigma(u) \quad \text{for all } \alpha \geq 0, u \in \mathbb{R}.$$

In particular, σ satisfies this property. This assumption is used for Fisher–Kibble decoupling (Lemma 9) and for separating radial and angular fluctuations in the three-layer kernel.

B.4. EOC scaling and the origin of $1/r$

Let $\Sigma_w = \text{Cov}(w_j^{(\ell)})$. Writing $q^{(\ell)} = \mathbb{E}\|h^{(\ell)}(x)\|^2$ for a fixed unit-norm input, weight independence yields the variance recursion

$$q^{(\ell)} = \frac{\sigma_A^2}{2} q^{(\ell-1)} \text{Tr}(\Sigma_w),$$

The edge-of-chaos (EOC) condition chooses σ_A so that the linearized coefficient equals 1,

$$\frac{\sigma_A^2}{2} \text{Tr}(\Sigma_w) = 1,$$

preventing exploding/vanishing signal propagation across depth. In particular, under $\Sigma_w = I_{\mathbf{d}_{\ell-1}}/\mathbf{d}_{\ell-1}$ one has $\text{Tr}(\Sigma_w) = 1$ and $\sigma_A = \sqrt{2}$.

In the bottleneck regime $\mathbf{d}_{\ell-1} = r$ for $\ell \geq 2$, EOC prescribes $\Sigma_w = I_r/r$. For ReLU, the (normalized) derivative kernel has the closed form

$$\bar{\Sigma}^{(\ell)}(\rho) = \mathbb{P}(Z > 0, Z' > 0) = \frac{1}{2} - \frac{\arccos(\rho)}{2\pi} \in \left[0, \frac{1}{2}\right],$$

so $\dot{\Sigma}^{(\ell)}(x, x') = \bar{\Sigma}^{(\ell)}(\rho_\ell)$ remains $O(1)$ under EOC and is uniformly bounded by $1/2$. The RF-LR NTK recursion (Theorem 3) carries an explicit prefactor $1/r$ at bottleneck layers because gradients propagate through an r -dimensional bottleneck and the corresponding Jacobian metric concentrates on an r -dimensional subspace; see Appendix B.6.1 for the detailed derivation.

B.5. Finite-width effects

Beyond the infinite-width limit, one can in principle study finite-width corrections to the kernel via cumulant/resolvent expansions or diagrammatic methods; see Dyer and Gur-Ari (2019) for a Feynman-diagram approach in wide-network asymptotics. These corrections are deferred to future work.

B.6. Proofs of NTK Recursions

B.6.1. PROOF OF THEOREM 3: INFINITE-WIDTH NTK RECURSION

Proof sketch. Decompose the NTK at layer ℓ into (i) a bias contribution (constant +1), (ii) Term 2: gradient inner products for the layer- ℓ readouts $A^{(\ell)}$, and (iii) Term 1: backpropagated NTK from previous layers times the Jacobian metric. Apply the Law of Large Numbers conditionally on $\mathbf{h}^{(\ell-1)}$ as $n_\ell \rightarrow \infty$; the sequential limit ensures lower layers have already converged.

Proof We prove the recursion by decomposing the NTK at layer ℓ into contributions of the trainable parameters up to layer ℓ , and invoking the Law of Large Numbers (LLN) under the sequential infinite-width limit. Let $f^{(\ell)}$ denote the network output truncated at layer ℓ . For finite width, the (random) NTK reads

$$\Theta_{\text{rand}}^{(\ell)}(x, x') = \underbrace{(\nabla_{\theta^{(\ell-1)}} f^{(\ell)}(x))^\top (\nabla_{\theta^{(\ell-1)}} f^{(\ell)}(x'))}_{\text{Term 1}} + \underbrace{\sum_{j=1}^{n_\ell} \frac{\partial f^{(\ell)}(x)}{\partial A_j^{(\ell)}} \frac{\partial f^{(\ell)}(x')}{\partial A_j^{(\ell)}}}_{\text{Term 2}}. \quad (13)$$

Bias parameters. We include a trainable layer bias $c^{(\ell)}$ (initialized at zero). The gradient feature with respect to $c^{(\ell)}$ is constant, so it contributes an additive constant term to the NTK; under our normalization this contribution is 1. This yields the additive +1 term in the recursion.

Term 2 (layer- ℓ output weights $A^{(\ell)}$). By the forward pass, $\partial f^{(\ell)}(x)/\partial A_j^{(\ell)} = \frac{\sigma_A}{\sqrt{n_\ell}} \sigma(\mathbf{w}_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x))$. Therefore

$$\text{Term 2} = \frac{\sigma_A^2}{n_\ell} \sum_{j=1}^{n_\ell} \sigma(\mathbf{w}_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x)) \sigma(\mathbf{w}_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x')). \quad (14)$$

Conditionally on $\mathbf{h}^{(\ell-1)}$, the summands are i.i.d. over $\mathbf{w}_j^{(\ell)}$. As $n_\ell \rightarrow \infty$ (with lower layers already taken to their limits), LLN yields

$$\text{Term 2} \xrightarrow{p} \sigma_A^2 \Sigma^{(\ell)}(x, x'). \quad (15)$$

Term 1 (previous layers). By the chain rule,

$$\nabla_{\theta^{(\ell-1)}} f^{(\ell)}(x) = \left(\frac{\partial f^{(\ell)}(x)}{\partial \mathbf{h}^{(\ell-1)}(x)} \right) \nabla_{\theta^{(\ell-1)}} \mathbf{h}^{(\ell-1)}(x). \quad (16)$$

Hence Term 1 factorizes as $\Theta_{\text{rand}}^{(\ell-1)}(x, x')$ multiplied by a Jacobian metric:

$$\text{Term 1} = \Theta_{\text{rand}}^{(\ell-1)}(x, x') \cdot \left\langle \frac{\partial f^{(\ell)}}{\partial \mathbf{h}^{(\ell-1)}}(x), \frac{\partial f^{(\ell)}}{\partial \mathbf{h}^{(\ell-1)}}(x') \right\rangle. \quad (17)$$

Taking the inner product and applying LLN as $n_\ell \rightarrow \infty$ yields

$$\left\langle \frac{\partial f^{(\ell)}}{\partial \mathbf{h}^{(\ell-1)}}(x), \frac{\partial f^{(\ell)}}{\partial \mathbf{h}^{(\ell-1)}}(x') \right\rangle \xrightarrow{p} \sigma_A^2 \dot{\Sigma}^{(\ell)}(x, x'). \quad (18)$$

Therefore,

$$\text{Term 1} \xrightarrow{p} \Theta^{(\ell-1)}(x, x') \cdot \sigma_A^2 \dot{\Sigma}^{(\ell)}(x, x'). \quad (19)$$

Combining the bias contribution with the two limits completes the recursion. For bottleneck layers $\ell \geq 2$, the Jacobian and output contributions scale with $1/r$ (from the r -dimensional bottleneck); we absorb σ_A^2 into the normalization. Thus

$$\Theta^{(1)}(x, x') = 1 + \Sigma^{(1)}(x, x'), \quad (20)$$

$$\Theta^{(\ell)}(x, x') = 1 + \frac{1}{r} \Theta^{(\ell-1)}(x, x') \cdot \sigma_A^2 \dot{\Sigma}^{(\ell)}(x, x') + \frac{1}{r} \sigma_A^2 \Sigma^{(\ell)}(x, x') \quad (\ell \geq 2). \quad (21)$$

The sequential limit assumption ensures that lower-layer random objects have already converged when taking the next-layer limit, justifying the conditional LLN applications. $\blacksquare$

B.6.2. PROOF OF THEOREM 2: GAUSSIAN PROCESS COMPOSITION

Proof sketch. By induction on ℓ : for $\ell = 0$ the output is deterministic; for $\ell \geq 1$, conditionally on $\mathbf{h}^{(\ell-1)}$ the forward pass is a sum of i.i.d. terms over j , so the multivariate CLT yields Gaussian convergence. The covariance matches the base kernel $\Sigma^{(\ell)}$ by definition.

Proof We prove by induction on ℓ that $\mathbf{h}^{(\ell)}$ converges to a Gaussian process with covariance kernel $\Sigma^{(\ell)}$.

Base case $\ell = 0$. For $\ell = 0$, we have $\mathbf{h}^{(0)}(x) = x$, which is deterministic and trivially a Gaussian process.

Inductive step. Assume that $\mathbf{h}^{(\ell-1)}$ converges to a Gaussian process with covariance kernel $\Sigma^{(\ell-1)}$. Consider the forward pass at layer ℓ :

$$\mathbf{h}^{(\ell)}(x) = \frac{1}{\sqrt{n_\ell}} \sum_{j=1}^{n_\ell} A_j^{(\ell)} \sigma\left(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x)\right) + c^{(\ell)}. \quad (22)$$

For any finite collection of inputs $\{x_1, \dots, x_m\}$, conditionally on $\mathbf{h}^{(\ell-1)}$, the terms $A_j^{(\ell)} \sigma\left(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x_i)\right)$ are independent across j and identically distributed. Since $A_j^{(\ell)} \sim \mathcal{N}(0, \sigma_A^2)$ and $w_j^{(\ell)}$ are frozen Gaussian draws, the conditional distribution of each term is centered with finite variance (and $c^{(\ell)}$ is deterministic at initialization).

By the multivariate Central Limit Theorem, as $n_\ell \rightarrow \infty$, the vector $(\mathbf{h}^{(\ell)}(x_1), \dots, \mathbf{h}^{(\ell)}(x_m))$ converges in distribution to a multivariate Gaussian. The covariance is given by

$$\mathbb{E}\left[\mathbf{h}^{(\ell)}(x_i) \mathbf{h}^{(\ell)}(x_j)\right] = \mathbb{E}_w\left[\sigma\left(w^\top \mathbf{h}^{(\ell-1)}(x_i)\right) \sigma\left(w^\top \mathbf{h}^{(\ell-1)}(x_j)\right)\right] = \Sigma^{(\ell)}(x_i, x_j), \quad (23)$$

where the expectation is taken over the frozen weights w and the limiting Gaussian process $\mathbf{h}^{(\ell-1)}$. The last equality follows from the definition of the base kernel in Definition 1.

The sequential infinite-width limit ensures that $\mathbf{h}^{(\ell-1)}$ has already converged to its Gaussian process limit when taking the limit for layer ℓ , justifying the application of the CLT conditionally on $\mathbf{h}^{(\ell-1)}$. $\blacksquare$

B.6.3. PROOF OF THREE-LAYER NTK COROLLARY

Proof sketch. Apply the recursion (Theorem 3) for $\ell = 1$ and $\ell = 2$; for $\ell = 1$ use $\Theta^{(0)} = 0$; for $\ell = 2$ substitute $\Theta^{(1)} = 1 + \Sigma^{(1)}$. Under homogeneity, the mean form involves expectations over Fisher (angular) and Kibble (radial); as $r \rightarrow \infty$ these concentrate, recovering K_∞ .

Proof Apply Theorem 3 for $\ell = 1$ and $\ell = 2$. For $\ell = 1$, since $\Theta^{(0)} = 0$, the recursion gives:

$$\Theta^{(1)}(x, x') = 1 + \Theta^{(0)}(x, x') \cdot \dot{\Sigma}^{(1)}(x, x') + \Sigma^{(1)}(x, x') = 1 + \Sigma^{(1)}(x, x'). \quad (24)$$

For $\ell = 2$, substitute $\Theta^{(1)} = 1 + \Sigma^{(1)}$ into the recursion:

$$\Theta^{(2)}(x, x') = 1 + \frac{1}{r} \Theta^{(1)}(x, x') \cdot \dot{\Sigma}^{(2)}(x, x') + \frac{1}{r} \Sigma^{(2)}(x, x'). \quad (25)$$

This is the stated three-layer recursion with the bias contribution.

Mean explicit form (low-rank). Under the homogeneous ReLU setting on the sphere, the mean three-layer NTK (with $1/r$ factors) is

$$\tilde{\Theta}^{(2)}(\rho) = 1 + \frac{1}{r} \Theta^{(1)}(\rho) \mathbb{E}\left[1 - \frac{\arccos(\hat{\rho}_r)}{\pi}\right] + \frac{1}{r} \mathbb{E}\left[\Sigma^{(1)}(\hat{\rho}_r) \|x_1\| \|y_1\|\right], \quad (26)$$

where the expectations are over Fisher and Kibble. As $r \rightarrow \infty$, $\hat{\rho}_r \rightarrow \rho$ and $\|x_1\| \|y_1\|/r \rightarrow 1$, so the limit $K_\infty(\rho) = \Theta^{(1)}(\rho)(1 - \arccos(\rho)/\pi) + \Sigma^{(1)}(\rho) + 1$ is recovered (without the $1/r$ prefactors, since the low-rank scaling is absorbed in the limit). $\blacksquare$

B.6.4. PROOF OF GENERAL L -LAYER NTK COROLLARY

Remark 14 (Explicit $2L$ -term unrolling) *Equivalently, expanding the recursion yields an explicit linear number of terms (namely $2L$). For example, for $L = 3$:*

$$\begin{aligned}\Theta^{(3)}(x, x') &= 1 + \frac{1}{r} \dot{\Sigma}^{(2)} \cdot \dot{\Sigma}^{(3)} + \frac{1}{r} \dot{\Sigma}^{(3)} \\ &\quad + \frac{1}{r^2} \Sigma^{(1)} \cdot \dot{\Sigma}^{(2)} \cdot \dot{\Sigma}^{(3)} + \frac{1}{r} \Sigma^{(2)} \cdot \dot{\Sigma}^{(3)} + \frac{1}{r} \Sigma^{(3)}.\end{aligned}\quad (27)$$

Each bottleneck layer contributes a factor $1/r$; layer 1 has no such factor. In total, there are $2L$ terms for depth L .

Proof sketch. By induction on L : for $L = 1$ the recursion gives $\Theta^{(1)} = 1 + \Sigma^{(1)}$. For $L \geq 2$, expand $\Theta^{(L)}$ via the recursion into a bias term and two paths (derivative-propagated and fresh-base); each path contributes L terms with the appropriate $1/r$ factors, yielding $2L$ terms in total.

Proof We prove the explicit form by induction on L , accounting for the $+1$ bias terms in the recursion.

Base case $L = 1$. From Theorem 3, $\Theta^{(1)} = 1 + \Sigma^{(1)}$. The formula with $L = 1$ gives:

$$\Theta^{(1)}(x, x') = \sum_{\ell=1}^1 \left(\prod_{k=\ell+1}^1 \dot{\Sigma}^{(k)}(x, x') \right) + \sum_{\ell=1}^1 \Sigma^{(\ell)}(x, x') \prod_{k=\ell+1}^1 \dot{\Sigma}^{(k)}(x, x'). \quad (28)$$

For the first sum: when $\ell = 1$, $\prod_{k=2}^1 \dot{\Sigma}^{(k)} = 1$ (empty product). For the second sum: when $\ell = 1$, $\Sigma^{(1)} \cdot \prod_{k=2}^1 \dot{\Sigma}^{(k)} = \Sigma^{(1)} \cdot 1 = \Sigma^{(1)}$. Therefore, $\Theta^{(1)} = 1 + \Sigma^{(1)}$, which matches the recursion.

Inductive step. Assume the formula holds for depth $L - 1$:

$$\Theta^{(L-1)}(x, x') = \sum_{\ell=1}^{L-1} \left(\prod_{k=\ell+1}^{L-1} \dot{\Sigma}^{(k)}(x, x') \right) + \sum_{\ell=1}^{L-1} \Sigma^{(\ell)}(x, x') \prod_{k=\ell+1}^{L-1} \dot{\Sigma}^{(k)}(x, x'). \quad (29)$$

By Theorem 3, for $L \geq 2$,

$$\Theta^{(L)}(x, x') = 1 + \frac{1}{r} \Theta^{(L-1)}(x, x') \cdot \dot{\Sigma}^{(L)}(x, x') + \frac{1}{r} \Sigma^{(L)}(x, x'). \quad (30)$$

Substituting the inductive hypothesis and distributing the $1/r$ factor yields the stated closed form; the first sum gains a new term 1 (from $\ell = L$) and each existing term is multiplied by $(1/r)\dot{\Sigma}^{(L)}$, while the second sum gains $(1/r)\Sigma^{(L)}$ and each existing term is multiplied by $(1/r)\dot{\Sigma}^{(L)}$. The first term 1 corresponds to $\ell = L$ in the first sum (with empty product $\prod_{k=L+1}^L \dot{\Sigma}^{(k)} = 1$). The last term $\Sigma^{(L)}$ corresponds to $\ell = L$ in the second sum (with empty product $\prod_{k=L+1}^L \dot{\Sigma}^{(k)} = 1$). Therefore:

$$\Theta^{(L)}(x, x') = \sum_{\ell=1}^L \left(\prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x') \right) + \sum_{\ell=1}^L \Sigma^{(\ell)}(x, x') \prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x'), \quad (31)$$

completing the induction. The total number of terms is $L + L = 2L$, which grows linearly with depth. ■

B.6.5. REFERENCE: FULL-WIDTH MLP LIMITING NTK AT THE EDGE OF CHAOS

Proposition 15 (Closed form for full-width MLP EOC NTK) *Let $l \geq 1$ and consider an infinitely wide l -layer MLP with (a, b) -ReLU activation initialized at the edge of chaos as in [Terjék and González-Sánchez \(2025\)](#). Denote by ϱ the corresponding cosine map and by $\rho_1(x, x')$ the input cosine similarity. Then the limiting NTK satisfies*

$$K_\infty(x, x') = \|x\| \|x'\| \left(\sum_{k=1}^l \varrho^{\circ(k-1)}(\rho_1(x, x')) \prod_{k'=k}^{l-1} \varrho'(\varrho^{\circ(k'-1)}(\rho_1(x, x'))) \right) I_{m_l},$$

with the convention that the empty product (for $k = l$) equals 1. See [Terjék and González-Sánchez \(2025\)](#) for the derivation.

B.7. Comparison with the classical fully-trained NTK recursion

For comparison, consider a standard fully-connected MLP where *all* weights (and biases) are trained under NTK parameterization. In the sequential infinite-width limit, the classical NTK recursion takes the form [Jacot et al. \(2018\)](#)

$$\Theta_{\text{MLP}}^{(0)}(x, x') = 0, \quad \Theta_{\text{MLP}}^{(\ell)}(x, x') = \Theta_{\text{MLP}}^{(\ell-1)}(x, x') \dot{K}^{(\ell)}(x, x') + K^{(\ell)}(x, x'), \quad \ell \geq 1, \quad (32)$$

where $K^{(\ell)}$ is the (deterministic) NNGP kernel at layer ℓ and $\dot{K}^{(\ell)}$ the associated derivative kernel (both obtained from the signal-propagation recursion). Unrolling (32) yields one contribution per layer (a sum of L terms, each propagated through subsequent $\dot{K}^{(k)}$ factors).

In contrast, the RF-LR recursion of Theorem 3 (i) carries explicit $1/r$ prefactors at bottleneck layers, and (ii) uses conditional base/derivative kernels $\Sigma^{(\ell)}, \dot{\Sigma}^{(\ell)}$ rather than the classical NNGP kernels $K^{(\ell)}, \dot{K}^{(\ell)}$. These differences lead to the $2L$ -term expansion in Corollary 4, reflecting separate bias-path and fresh-basis contributions, each propagated through subsequent derivative kernels and weighted by bottleneck factors.

Relation to fully trained MLPs at the edge of chaos. For fully trained (full-width) MLPs at the edge of chaos, the limiting NTK admits a single-layer sum structure (one contribution per layer) without RF-LR bottleneck prefactors; see Proposition 15 for a reference closed form [Terjék and González-Sánchez \(2025\)](#). In contrast, the RF-LR expansion (7) contains $2L$ terms because the recursion has both bias-path and fresh-basis-path contributions, each propagated through subsequent derivative kernels and weighted by explicit $1/r$ factors at bottlenecks.

B.8. Probabilistic recursion, GP concatenation, and bottleneck scaling

Random correlation chain and Fisher’s law. At each layer $\ell \geq 2$, the pair $(\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x'))$ is a $2r$ -dimensional Gaussian conditional on lower-layer randomness. The *layer- ℓ sample correlation*

$$\rho_\ell = \cos \angle(\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x')) = \frac{\langle \mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x') \rangle}{\|\mathbf{h}^{(\ell-1)}(x)\| \|\mathbf{h}^{(\ell-1)}(x')\|}$$

is a *random variable*. Given the population correlation $\rho = \varrho(\rho_{\ell-1})$ (the conditional mean under the ReLU cosine map), Fisher’s law [Fisher \(1915\)](#); [Hotelling \(1953\)](#) and Lemma 20 give $\mathbb{E}[(\rho_\ell - \rho)^2] \leq C/r$ and $\mathbb{P}(|\rho_\ell - \rho| \geq t) \leq 6 \exp(-c r t^2)$. Thus the chain $\rho_1 \rightarrow \rho_2 \rightarrow \dots \rightarrow \rho_L$ is a *Markov*

chain of random variables with $O(1/\sqrt{r})$ fluctuations at each transition. The NTK recursion passes through these random correlations successively: $\Sigma^{(\ell)} = \bar{\Sigma}^{(\ell)}(\rho_\ell)$, $\dot{\Sigma}^{(\ell)} = \bar{\dot{\Sigma}}^{(\ell)}(\rho_\ell)$ with random ρ_ℓ . The deterministic proxy (Definition D.4) replaces this random path by the deterministic iterates $\rho_k = \varrho^{(k-1)}(\rho_1)$, which represent the mean/infinite- r limit path.

Kernel randomness and GP concatenation. For $\ell > 1$, $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$ are random fields because they depend on $\mathbf{h}^{(\ell-1)}$, which is the output of a (conditional) Gaussian process. By Theorem 2, the hidden states form a composition of GPs: each layer's output is conditionally Gaussian given the previous layer, but the unconditional law is a deep GP. In the bottleneck regime, the pair $(\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x'))$ is a $2r$ -dimensional Gaussian conditional on lower-layer randomness, with covariance determined by $\Sigma^{(\ell-1)}$. The base kernel $\Sigma^{(\ell)}$ and derivative kernel $\dot{\Sigma}^{(\ell)}$ are thus expectations over the frozen features $\mathbf{w}^{(\ell)}$, conditional on this Gaussian pair, and inherit randomness through $\Sigma^{(\ell-1)}$.

Bottleneck scaling and explicit ReLU bounds. At a bottleneck layer $\ell \geq 2$, the RF-LR NTK recursion carries an explicit prefactor $1/r$ (Theorem 3). For ReLU and isotropic weights with $\text{Cov}(w) = I_r/r$, one has

$$\dot{\Sigma}^{(\ell)}(x, x') = \mathbb{E}_w \left[\dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x)) \dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x')) \|w\|^2 \right] = \bar{\dot{\Sigma}}^{(\ell)}(\rho_\ell),$$

where $\rho_\ell = \cos \angle(\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x'))$ and $\bar{\dot{\Sigma}}^{(\ell)}(\rho)$ denotes the corresponding scalar derivative kernel evaluated at cosine similarity $\rho \in [-1, 1]$. Under the EOC normalization $\text{Tr}(\text{Cov}(w)) = 1$, the ReLU derivative kernel has the explicit form

$$\bar{\dot{\Sigma}}^{(\ell)}(\rho) = \mathbb{P}(Z > 0, Z' > 0) = \frac{1}{2} - \frac{\arccos(\rho)}{2\pi} \in \left[0, \frac{1}{2}\right],$$

where (Z, Z') is centered Gaussian with correlation ρ . In particular, $\sup_{\rho \in [-1, 1]} \bar{\dot{\Sigma}}^{(\ell)}(\rho) = 1/2$, uniformly in ℓ . Similarly, $\Sigma^{(\ell)}(x, x') = \mathbb{E}_w [\sigma(w^\top \mathbf{h}^{(\ell-1)}(x)) \sigma(w^\top \mathbf{h}^{(\ell-1)}(x'))]$ is an $O(1)$ scalar kernel, and the $1/r$ factors enter through the recursion.

Probabilistic recursion (nested expectations). The mean (probabilistic) NTK $\tilde{\Theta}^{(L)} = \mathbb{E}[\Theta^{(L)}]$ satisfies a recursion involving expectations of products:

$$\tilde{\Theta}^{(1)} = 1 + \mathbb{E}[\Sigma^{(1)}], \quad \tilde{\Theta}^{(\ell)} = 1 + \frac{1}{r} \mathbb{E}[\Theta^{(\ell-1)} \cdot \dot{\Sigma}^{(\ell)}] + \frac{1}{r} \mathbb{E}[\Sigma^{(\ell)}], \quad \ell \geq 2. \quad (33)$$

To emphasize the nested structure, write $\mathbb{E}_{\leq \ell-1}$ for expectation over all randomness up to depth $\ell-1$, and $\mathbb{E}_{w^{(\ell)}}[\cdot | \mathbf{h}^{(\ell-1)}]$ for expectation over the frozen feature direction $w^{(\ell)}$ at layer ℓ , conditional on the previous hidden states. Then the mean kernels are iterated conditional expectations:

$$\begin{aligned} \mathbb{E}[\Sigma^{(\ell)}(x, x')] &= \mathbb{E}_{\leq \ell-1} \left[\mathbb{E}_{w^{(\ell)}} [\sigma(w^\top \mathbf{h}^{(\ell-1)}(x)) \sigma(w^\top \mathbf{h}^{(\ell-1)}(x')) | \mathbf{h}^{(\ell-1)}] \right], \\ \mathbb{E}[\Theta^{(\ell-1)}(x, x') \dot{\Sigma}^{(\ell)}(x, x')] &= \mathbb{E}_{\leq \ell-1} \left[\Theta^{(\ell-1)}(x, x') \cdot \mathbb{E}_{w^{(\ell)}} [\dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x)) \dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x')) \|w\|^2 | \mathbf{h}^{(\ell-1)}] \right]. \end{aligned}$$

Since $\Theta^{(\ell-1)}$ and $\Sigma^{(\ell)}, \dot{\Sigma}^{(\ell)}$ share the randomness of $\mathbf{h}^{(\ell-1)}$, one has $\mathbb{E}[\Theta^{(\ell-1)} \cdot \dot{\Sigma}^{(\ell)}] \neq \tilde{\Theta}^{(\ell-1)} \cdot \mathbb{E}[\dot{\Sigma}^{(\ell)}]$ in general; controlling this dependence (and the resulting nested conditional expectations) is the main technical point for a rigorous deep mean-kernel theory.

Exponential depth suppression (ReLU, bottlenecks). Assume for simplicity that all layers $\ell \geq 2$ are bottlenecks with the same rank r . For ReLU under EOC we have the explicit uniform bound

$$\bar{\Sigma}^{(\ell)}(\rho) \leq c_0 := \frac{1}{2}, \quad \forall \rho \in [-1, 1], \quad \forall \ell \geq 2.$$

From the closed form (7), the contribution from layer ℓ to $\Theta^{(L)}$ is multiplied by $\prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}$. Writing $\dot{\Sigma}^{(k)} = \bar{\Sigma}^{(k)}(\rho_k)$, we obtain

$$\frac{1}{r^{L-\ell}} \prod_{k=\ell+1}^L \dot{\Sigma}^{(k)} \leq \frac{1}{r^{L-\ell}} \prod_{k=\ell+1}^L \bar{\Sigma}^{(k)}(\rho_k) \leq \frac{c_0^{L-\ell}}{r^{L-\ell}}.$$

Hence, contributions from early layers ($\ell \ll L$) decay exponentially in $L - \ell$ when $c_0/r < 1$, and the NTK becomes increasingly localized to the top of the network as depth grows.

Proposition 16 (Effective depth decomposition and a $\log r$ window) *Assume the uniform derivative-kernel bound $\dot{\Sigma}^{(\ell)}(x, x') \leq c_0$ for all $\ell \geq 2$ and all input pairs (x, x') , and assume that for a given (x, x') one has a uniform base-kernel bound*

$$|\Sigma^{(\ell)}(x, x')| \leq C_\Sigma(x, x'), \quad \forall \ell \in \{1, \dots, L\}.$$

Fix $L \geq 2$, $r > c_0$, and an integer $m \in \{1, \dots, L\}$. Define the top- m truncation $\Theta_{\text{top},m}^{(L)}(x, x')$ by restricting both sums in the explicit expansion (7) to indices $\ell \in \{L - m + 1, \dots, L\}$. Then

$$|\Theta^{(L)}(x, x') - \Theta_{\text{top},m}^{(L)}(x, x')| \leq \frac{1 + C_\Sigma(x, x')}{1 - c_0/r} \left(\frac{c_0}{r}\right)^m.$$

In particular, taking $m = \lceil \alpha \log r \rceil$ (for any $\alpha > 0$) yields a $\log r$ effective-depth window: the contribution of layers $\ell \leq L - \lceil \alpha \log r \rceil$ is at most

$$|\Theta^{(L)}(x, x') - \Theta_{\text{top}, \lceil \alpha \log r \rceil}^{(L)}(x, x')| \leq \frac{1 + C_\Sigma(x, x')}{1 - c_0/r} \left(\frac{c_0}{r}\right)^{\lceil \alpha \log r \rceil} = o(r^{-p})$$

for every fixed $p > 0$ as $r \rightarrow \infty$.

Proof sketch. The remainder is the sum of terms in (7) with $\ell \leq L - m$; each such term carries a factor $\prod_{k=\ell+1}^L \dot{\Sigma}^{(k)} \leq c_0^{L-\ell}$ and hence $(c_0/r)^{L-\ell}$. Summing over $\ell \leq L - m$ yields a geometric-series tail; substituting $m = \lceil \alpha \log r \rceil$ gives the stated decay.

Proof By definition of $\Theta_{\text{top},m}^{(L)}$, the remainder is the sum of the terms in (7) with $\ell \leq L - m$. For such ℓ , we have $L - \ell \geq m$, and by the derivative-kernel bound $\dot{\Sigma}^{(k)}(x, x') \leq c_0$ we obtain

$$\frac{1}{r^{L-\ell}} \prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x') \leq \left(\frac{c_0}{r}\right)^{L-\ell}.$$

Moreover, for $\ell \leq L - m \leq L - 1$ the base-kernel term carries the same prefactor $r^{-(L-\ell)}$ in (7), hence

$$\frac{1}{r^{L-\ell}} |\Sigma^{(\ell)}(x, x')| \prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x') \leq C_\Sigma(x, x') \left(\frac{c_0}{r}\right)^{L-\ell}.$$

Summing these bounds over $\ell \leq L - m$ and using a geometric-series tail gives

$$|\Theta^{(L)}(x, x') - \Theta_{\text{top}, m}^{(L)}(x, x')| \leq (1 + C_\Sigma(x, x')) \sum_{j=m}^{\infty} \left(\frac{c_0}{r}\right)^j = \frac{1 + C_\Sigma(x, x')}{1 - c_0/r} \left(\frac{c_0}{r}\right)^m,$$

as claimed. The $\log r$ specialization follows by substituting $m = \lceil \alpha \log r \rceil$ and observing that $(c_0/r)^{\lceil \alpha \log r \rceil} = o(r^{-p})$ for every fixed $p > 0$. $\blacksquare$

B.9. Comparison table: depth/rank effects in the kernel regime

	MLP at EOC Terjék and González-Sánchez (2025)	RF-LR (this work)
Trainable	all weight layers	readouts $A^{(\ell)}$, biases $c^{(\ell)}$; frozen $w^{(\ell)}$
Scaling	depth L , Δ_ϕ	depth L , bottleneck r
Kernel magnitude	uncentered scale can grow with L	mean recursion saturates in L (Thm. 5)
Centered scale	correlation propagation	gap $\Theta^{(L)}(1) - \Theta^{(L)}(\rho) \asymp 1/(rL)$ (Thm. 5)

Table 1: Depth/rank effects in the kernel regime. MLP: fully trained behavior; RF-LR: explicit $1/r$ bottleneck, saturation and early-layer suppression.

B.10. Techniques and proof sketch

We highlight the main proof ingredients and how they combine.

Recursion and closed form. The RF-LR NTK recursion (Theorem 3) follows from a layer-wise decomposition of the gradient inner product into bias, readout, and backpropagated contributions, together with conditional law of large numbers arguments under the sequential infinite-width limit [Jacot et al. \(2018\)](#). Iterating the recursion yields the explicit $2L$ -term expansion (Corollary 4).

Microscopic decoupling and concentration. For three layers, the empirical kernel depends on the sample cosine and norm product of rank- r Gaussian projections. Rotational invariance yields Fisher–Kibble decoupling (Lemma 9), separating angular and radial fluctuations [Fisher \(1915\)](#); [Hotelling \(1953\)](#); [Kibble \(1941\)](#). Concentration of Gaussian norms and sample correlations then yields sub-Gaussian deviations in r for the empirical kernel (Theorem 25 and Corollary 26).

RKHS identification via endpoint expansions. The RKHS of zonal kernels on the sphere is controlled by the Puiseux behavior near $\rho = \pm 1$ [Bietti and Bach \(2021\)](#). We compute the mean three-layer kernel’s endpoint expansion by combining Fisher’s density with a hypergeometric connection formula, showing that the leading $t^{1/2}$ exponent is preserved after taking expectations, hence the mean kernel induces the same RKHS as the shallow ReLU kernel (Corollary 13).

Depth scaling and effective depth. Depth dependence is controlled by correlation propagation and by products of derivative kernels. In the bottleneck regime, the RF-LR recursion carries explicit $1/r$ prefactors at each bottleneck layer, and for ReLU one has $\sup_\rho \bar{\Sigma}^{(\ell)}(\rho) \leq 1/2$, which yields exponential suppression of early-layer contributions as depth increases (Section 4). A refined mean-kernel statement is obtained by analyzing the deterministic proxy recursion (Theorem 5).

Future work: quadratic random-matrix scaling. Establishing bulk spectral laws for the *empirical* RF-LR NTK in a quadratic scaling regime $n \rightarrow \infty$ requires additional resolvent/local-law inputs. The Benigni–Paquette framework [Benigni and Paquette \(2025\)](#) establishes RMT limits for Gram spectra in the extensive-width regime ($n \sim N^2$) with layers taken in the RMT limit, via resolvent replacement and Gaussian-equivalence machinery for shallow two-layer NTKs. Adapting these methods to the present low-rank random-feature structure and to depth $L \geq 2$ is left for future work.

Appendix C. Proofs of low-rank NTK RKHS

C.1. Background and statements for Section 5

RKHS viewpoint in the NTK regime. In the NTK regime, training dynamics linearize:

$$f_t(x) \approx f_0(x) + \langle \nabla_{\theta} f_0(x), \theta_t - \theta_0 \rangle.$$

The learned function stays in the RKHS induced by the (limiting) kernel Θ , with norm

$$\|f\|_{\mathcal{H}_{\Theta}}^2 = \langle f, \Theta^{-1} f \rangle_{L^2}.$$

For RF-LR with frozen random features, universal approximation holds due to full support of the frozen Gaussian draws $w^{(\ell)}$, so any function in the Barron space [E et al. \(2021\)](#) can be approximated with error $O(1/\sqrt{N})$. Classical NTK analysis suggests that feature weights move little during training; RF-LR enforces this by design by freezing the feature weights, which isolates kernel-regime behavior without an additional weight-deviation analysis.

Mean RKHS and random features. Each realization ω of the random features yields a kernel $K_{\omega}(x, x') = \Theta_{\omega}^{(2)}(x, x')$ and an induced RKHS $\mathcal{H}_{\omega}$. The mean kernel $\tilde{K}(x, x') = \mathbb{E}_{\omega}[K_{\omega}(x, x')]$ is deterministic; its RKHS $\mathcal{H}_{\tilde{K}}$ is the RKHS of $\tilde{K}$. It is not $\mathbb{E}[\mathcal{H}_{\omega}]$ (expectation of Hilbert spaces is undefined). For each ω , the feature map $\varphi_{\omega}(x) = K_{\omega}(\cdot, x) \in \mathcal{H}_{\omega}$ is random, whereas the canonical feature map of the mean kernel $\varphi_{\tilde{K}}(x) = \tilde{K}(\cdot, x) \in \mathcal{H}_{\tilde{K}}$ is deterministic. The mean RKHS describes the typical function space obtained after averaging over random feature realizations.

Remark (application of Bietti–Bach). Once the mean kernel’s Puiseux endpoint behavior is established (Appendix C.7), the RKHS equivalence in Corollary 13 follows directly from the characterization theorem of [Bietti and Bach \(2021\)](#).

Corollary 17 (Mean NTK over Fisher–Kibble has same RKHS: three-layer case) *Under Assumption 1 and the RF-LR setting with ReLU nonlinearity and isotropic random features on $\mathbb{S}^{d-1}$, the mean three-layer NTK $\tilde{\Theta}^{(2)}$ (obtained by taking expectations over Fisher and Kibble distributions) is a zonal kernel that induces the same RKHS as the shallow ReLU kernel. In particular, the RKHSs coincide as sets with equivalent norms.*

Proof: See Appendix C.7.

Main RKHS result (stated in the main text). Corollary 13 is stated in Section 5. It follows by combining the Puiseux endpoint analysis in Appendix C.7 with the RKHS characterization of [Bietti and Bach \(2021\)](#).

C.2. Discussion on Fisher and Kibble Distributions

Lemma 18 (Fisher–Kibble decoupling) *Let x, y be input vectors and let x_1, y_1 denote their rank- r random projections. Define the empirical correlation $\rho_1 = \cos \angle(x_1, y_1)$ and squared norms $u = \|x_1\|^2, v = \|y_1\|^2$. Then:*

- ρ_1 follows Fisher’s correlation distribution [Fisher \(1915\)](#) [Hotelling \(1953\)](#), centered at the true correlation ρ .
- (u, v) follow Kibble’s [Kibble \(1941\)](#) bivariate Gamma (chi-square) law.
- Angular and radial parts are independent: $p(\rho_1, u, v) = p_{\text{Fisher}}(\rho_1) p_{\text{Kibble}}(u, v)$.

Remark 19 (Fisher–Kibble decoupling and independence) *By rotational invariance, the empirical correlation and squared norms satisfy*

$$\rho_1 \sim \text{Fisher}(\rho, r), \quad (u, v) \sim \text{Kibble}(r, \rho), \quad \text{and} \quad p(\rho_1, u, v) = p_{\text{Fisher}}(\rho_1) p_{\text{Kibble}}(u, v).$$

Homogeneity and factorization (why angular and radial parts decouple). *For positively 1-homogeneous activations (Assumption 1), the first-layer base kernel separates radial and angular dependence: for all $x, x' \neq 0$,*

$$\Sigma^{(1)}(x, x') = \|x\| \|x'\| \tilde{\Sigma}^{(1)}(\rho), \quad \rho = \frac{\langle x, x' \rangle}{\|x\| \|x'\|},$$

where $\tilde{\Sigma}^{(1)}$ is a scalar function on $[-1, 1]$ (for ReLU it is the standard arc-cosine kernel). Under isotropic Gaussian projections, rotational invariance implies that the empirical correlation $\rho_1 = \cos \angle(x_1, y_1)$ depends only on the angular part of the projected pair, while the norms $(\|x_1\|, \|y_1\|)$ depend only on the radial part; consequently ρ_1 is independent of $(u, v) = (\|x_1\|^2, \|y_1\|^2)$ and the joint law factorizes as stated above. The Fisher distribution density [Fisher \(1915\)](#) is

$$p(\rho_1 \mid \rho, r) = \frac{(r-2) \Gamma(r-1) (1-\rho^2)^{\frac{r-1}{2}} (1-\rho_1^2)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r-\frac{1}{2}) (1-\rho\rho_1)^{r-\frac{3}{2}}} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r-\frac{1}{2}; \frac{1+\rho\rho_1}{2}\right), \quad r > 2.$$

For large r , the inverse Gudermann function applied to the complementary angle is approximately normal:

$$\text{gd}^{-1}\left(\frac{\pi}{2} - \arccos(\rho_1)\right) \approx \mathcal{N}\left(\text{arctanh}(\rho), \frac{1}{r-3}\right), \quad \text{hence} \quad \text{Var}(\rho_1) = O(1/r).$$

Kibble’s [Kibble \(1941\)](#) bivariate chi-square law for $(u, v) = (\|x_1\|^2, \|y_1\|^2)$ has density

$$f(u, v) = \frac{(uv)^{\frac{r/2-1}{2}} \exp\left(-\frac{u+v}{2(1-\rho^2)}\right)}{\Gamma(r/2) (2(1-\rho^2))^{\frac{r}{2}+1} \rho^{\frac{r/2-1}{2}}} I_{\frac{r}{2}-1}\left(\frac{\rho\sqrt{uv}}{1-\rho^2}\right), \quad u, v \geq 0,$$

where I_ν is the modified Bessel function of the first kind. The moments are

$$\mathbb{E}[u] = \mathbb{E}[v] = r, \quad \text{Var}(u) = \text{Var}(v) = 2r, \quad \text{Cov}(u, v) = 2r \rho^2.$$

Defining $w_r = \sqrt{uv}/r$, concentration results imply $\mathbb{E}[w_r] \rightarrow 1$ and $\text{Var}(w_r) = O(1/r)$.

Compact three-layer empirical NTK representation. *In the three-layer low-rank setting, denote*

$$\rho_1 = \cos \angle(\mathbf{h}^{(1)}(x), \mathbf{h}^{(1)}(x')), \quad w_r = \frac{\|\mathbf{h}^{(1)}(x)\| \|\mathbf{h}^{(1)}(x')\|}{r}.$$

Then the empirical three-layer NTK admits the compact form

$$\Theta^{(2)}(x, x') = 1 + \frac{1}{r} \Theta^{(1)}(\rho_1) \left(1 - \frac{\arccos(\rho_1)}{\pi}\right) + \frac{1}{r} w_r \Sigma^{(1)}(\rho_1),$$

with $\Sigma^{(1)}(u) = \frac{1}{\pi} (\sqrt{1-u^2} + u(1 - \arccos u))$.

Proof sketch. The rank- r projections $x_1 = Px$, $y_1 = Py$ form a ρ -correlated Gaussian pair. By rotational invariance, the empirical correlation $\rho_1 = \cos \angle(x_1, y_1)$ (angular part) and the squared norms $(u, v) = (\|x_1\|^2, \|y_1\|^2)$ (radial part) are independent. The angular part follows Fisher's distribution; the radial part follows Kibble's bivariate chi-square law.

Proof Let $x, y \in \mathbb{R}^d$ be fixed input vectors with correlation $\rho = \langle x, y \rangle / (\|x\| \|y\|)$. Consider a rank- r random projection matrix $P \in \mathbb{R}^{r \times d}$ with entries $P_{ij} \sim \mathcal{N}(0, 1/d)$ i.i.d., so that $x_1 = Px$ and $y_1 = Py$ are the projected vectors.

Since P has i.i.d. Gaussian entries, the projected vectors $x_1, y_1 \in \mathbb{R}^r$ form a ρ -correlated bivariate Gaussian pair. Specifically, for each component $i = 1, \dots, r$, the pairs (x_{1i}, y_{1i}) are i.i.d. with joint distribution

$$\begin{pmatrix} x_{1i} \\ y_{1i} \end{pmatrix} \sim \mathcal{N} \left(\mathbf{0}, \begin{pmatrix} \|x\|^2/d & \rho \|x\| \|y\|/d \\ \rho \|x\| \|y\|/d & \|y\|^2/d \end{pmatrix} \right). \quad (34)$$

By rotational invariance of the Gaussian projection and the fact that the correlation ρ is preserved under orthogonal transformations, we can assume without loss of generality that the covariance structure factors into angular and radial components.

Angular part (Fisher distribution). The empirical correlation is defined as

$$\rho_1 = \frac{\langle x_1, y_1 \rangle}{\|x_1\| \|y_1\|} = \frac{\sum_{i=1}^r x_{1i} y_{1i}}{\sqrt{\sum_{i=1}^r x_{1i}^2} \sqrt{\sum_{i=1}^r y_{1i}^2}}. \quad (35)$$

This is the sample correlation coefficient of r pairs of correlated Gaussian random variables. Fisher [Fisher \(1915\)](#); [Hotelling \(1953\)](#) showed that the distribution of ρ_1 depends only on the true correlation ρ and the sample size r , with density given in Remark 19. The key observation is that ρ_1 is a function purely of the angles between the projected vectors, independent of their magnitudes.

Radial part (Kibble distribution). The squared norms $u = \|x_1\|^2 = \sum_{i=1}^r x_{1i}^2$ and $v = \|y_1\|^2 = \sum_{i=1}^r y_{1i}^2$ are sums of correlated chi-square variables. Since (x_{1i}, y_{1i}) are ρ -correlated Gaussians, the bivariate distribution of (u, v) follows Kibble's [Kibble \(1941\)](#) bivariate chi-square law. The joint density involves the modified Bessel function $I_\nu(z)$, which appears because the dot product $\langle x_1, y_1 \rangle = \sum_{i=1}^r x_{1i} y_{1i}$ can be expressed as a weighted sum of products of correlated normals, whose distribution relates to Bessel functions through the generating function of the bivariate chi-square distribution.

Specifically, the joint characteristic function of (u, v) is

$$\phi(s, t) = \mathbb{E}[e^{i(su+tv)}] = (1 - 2i(1 - \rho^2)(s + t) - 4\rho^2 st)^{-r/2}, \quad (36)$$

and the inverse Fourier transform yields Kibble’s density, which contains the modified Bessel function $I_{\frac{r}{2}-1}$ as stated in Remark 19. The Bessel function arises from the integral representation of the bivariate chi-square density when the correlation $\rho \neq 0$.

Independence of angular and radial parts. Rotational invariance of isotropic Gaussian projections implies that $\rho_1 = \cos \angle(x_1, y_1)$ and the norms $(\|x_1\|, \|y_1\|)$ are independent, yielding the factorization $p(\rho_1, u, v) = p_{\text{Fisher}}(\rho_1) \cdot p_{\text{Kibble}}(u, v)$. This is the classical independence of the sample correlation from sample variances in bivariate normal data. $\blacksquare$

C.3. Open directions: deep mean NTK recursion and effective depth

The deterministic proxy (Definition D.4) analyzes the scalar recursion along the mean path $\rho_k = \varrho^{\circ(k-1)}(\rho_1)$. The actual network uses the *random* correlation chain ρ_ℓ with $O(1/\sqrt{r})$ per-layer fluctuations (Appendix B.8). A rigorous link requires: (i) propagating the Fisher-type concentration $\mathbb{E}[(\rho_\ell - \varrho^{\circ(\ell-1)}(\rho_1))^2] \leq C/r$ through depth to show $\Theta_{\text{random}}^{(L)} - \Theta_{\text{proxy}}^{(L)} = O_P(\sqrt{L/r})$ or similar; (ii) a rigorous recursion for $\tilde{\Theta}^{(\ell)}$ with explicit covariance bounds; (iii) extension of Fisher–Kibble decoupling to nested layers to control mixed terms such as $\mathbb{E}[\Theta^{(\ell-1)} \dot{\Sigma}^{(\ell)}]$ (where dependence through the deep GP chain prevents naive factorization). The exponential suppression of early-layer contributions suggests that very deep RF-LR may exhibit an “effective depth” phenomenon similar to ResNets, where only the top layers meaningfully contribute to the NTK.

Lemma 20 (Nonasymptotic concentration of Gaussian sample cosine) *Let $r \geq 1$ and let $(X, Y) \in \mathbb{R}^r \times \mathbb{R}^r$ have i.i.d. coordinates $\{(X_i, Y_i)\}_{i=1}^r$, each distributed as a centered bivariate normal with $\mathbb{E}[X_i^2] = \mathbb{E}[Y_i^2] = 1$ and $\mathbb{E}[X_i Y_i] = \rho \in (-1, 1)$. Define the empirical cosine (sample correlation without centering)*

$$\hat{\rho}_r = \frac{\langle X, Y \rangle}{\|X\| \|Y\|}.$$

Then there exist absolute constants $c, C > 0$ such that for all $t \in (0, 1)$,

$$\mathbb{P}(|\hat{\rho}_r - \rho| \geq t) \leq 6 \exp(-c r t^2), \quad \mathbb{E}[(\hat{\rho}_r - \rho)^2] \leq \frac{C}{r}.$$

Proof sketch. Write $\hat{\rho}_r = S_{xy} / \sqrt{S_{xx} S_{yy}}$ where S_{xx}, S_{yy}, S_{xy} are sample averages. Each of $S_{xx} - 1, S_{yy} - 1, S_{xy} - \rho$ is sub-exponential; Bernstein’s inequality yields tail bounds. On the event that S_{xx}, S_{yy} are close to 1, bound $|\hat{\rho}_r - \rho|$ via a ratio expansion; union bound over the three tails gives the stated rate.

Proof Write

$$S_{xx} = \frac{1}{r} \|X\|^2, \quad S_{yy} = \frac{1}{r} \|Y\|^2, \quad S_{xy} = \frac{1}{r} \langle X, Y \rangle, \quad \text{so that} \quad \hat{\rho}_r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}.$$

Each of $S_{xx} - 1, S_{yy} - 1$, and $S_{xy} - \rho$ is an average of i.i.d. centered sub-exponential random variables (since $X_i^2 - 1, Y_i^2 - 1$, and $X_i Y_i - \rho$ have finite ψ_1 norms for Gaussian data). Therefore,

by Bernstein's inequality, there exist absolute constants $c_0 > 0$ and $C_0 < \infty$ such that for all $t \in (0, 1)$,

$$\mathbb{P}(|S_{xx} - 1| \geq t) \leq 2e^{-c_0 r t^2}, \quad \mathbb{P}(|S_{yy} - 1| \geq t) \leq 2e^{-c_0 r t^2}, \quad \mathbb{P}(|S_{xy} - \rho| \geq t) \leq 2e^{-c_0 r t^2}.$$

On the event $E_t = \{|S_{xx} - 1| \leq t, |S_{yy} - 1| \leq t\}$ with $t \leq 1/2$, we have $S_{xx}, S_{yy} \in [1/2, 3/2]$, hence

$$\left| \frac{1}{\sqrt{S_{xx}S_{yy}}} - 1 \right| \leq C_0(|S_{xx} - 1| + |S_{yy} - 1|) \leq 2C_0 t$$

for an absolute constant C_0 . Using

$$\hat{\rho}_r - \rho = \frac{S_{xy} - \rho}{\sqrt{S_{xx}S_{yy}}} + \rho \left(\frac{1}{\sqrt{S_{xx}S_{yy}}} - 1 \right),$$

we obtain on E_t the bound $|\hat{\rho}_r - \rho| \leq C_1(|S_{xy} - \rho| + t)$ for an absolute constant C_1 . Taking $t \asymp t'$ and applying a union bound over the three Bernstein tails yields $\mathbb{P}(|\hat{\rho}_r - \rho| \geq t') \leq 6e^{-c r t'^2}$ for $t' \in (0, 1)$ and absolute $c > 0$. The second-moment bound follows by integrating the tail bound: $\mathbb{E}[(\hat{\rho}_r - \rho)^2] = \int_0^\infty 2u \mathbb{P}(|\hat{\rho}_r - \rho| \geq u) du \leq C/r$. $\blacksquare$

C.4. Concentration of the random Gram matrix around the proxy

Let $\hat{K}$ be the $n \times n$ random Gram matrix with entries $\hat{K}_{ij} = \hat{\Theta}_r^{(L)}(x_i, x_j)$ (the empirical RF-LR NTK over the random correlation chain), and let K_{proxy} be the deterministic proxy Gram matrix with entries $(K_{\text{proxy}})_{ij} = \Theta^{(L)}(\rho_{1,ij})$ (Definition D.4). Under the same per-layer sub-Gaussian concentration as in Appendix B.8,

$$\mathbb{E}[(\rho_\ell - \varrho^{\circ(\ell-1)}(\rho_1))^2] \leq C/r, \quad \mathbb{P}(|\rho_\ell - \varrho^{\circ(\ell-1)}(\rho_1)| \geq t) \leq 6 \exp(-c r t^2),$$

the kernel recursion is Lipschitz in the correlation path. Propagating the per-step bounds through L layers (e.g. by a union bound over the L steps and the fact that $\Theta^{(k)}$ is Lipschitz in ρ with constant $O(1)$), one obtains that for each pair (i, j) ,

$$|\hat{K}_{ij} - (K_{\text{proxy}})_{ij}| = O_P(\sqrt{L/r}).$$

A union over the n^2 pairs and the inequality $\|A\|_{\text{op}} \leq \|A\|_F \leq n \max_{i,j} |A_{ij}|$ yield the following. The bound in Proposition 21 is *sketch-based*; a full proof with explicit constants and rate is given only for the equicorrelated case (Theorem 7).

Proposition 21 (Proxy-empirical Gram matrix concentration) *Fix $n, L \geq 1, r > 1$, and inputs $x_1, \dots, x_n$ with pairwise cosine similarities $\rho_{1,ij} \in (-1, 1)$. Let $\hat{K}$ and K_{proxy} be as above. There exist constants $C, c > 0$ (depending on L, n and the $\rho_{1,ij}$) such that for any $\epsilon \in (0, 1)$,*

$$\mathbb{P}\left(\|\hat{K} - K_{\text{proxy}}\|_{\text{op}} \geq \epsilon\right) \leq C \exp\left(-c \frac{r \epsilon^2}{L n^2}\right).$$

In particular, for $r \rightarrow \infty$ with n, L fixed, $\|\hat{K} - K_{\text{proxy}}\|_{\text{op}} = o_P(1)$. If in addition the proxy Gram matrix has minimum eigenvalue $\lambda_{\min}(K_{\text{proxy}}|_{\mathbf{1}^\perp}) \geq \gamma > 0$, then for r large enough so that $\|\hat{K} - K_{\text{proxy}}\|_{\text{op}} < \gamma/2$ with high probability, Weyl's inequality gives

$$|\lambda_{\min}(\hat{K}|_{\mathbf{1}^\perp}) - \lambda_{\min}(K_{\text{proxy}}|_{\mathbf{1}^\perp})| \leq \gamma/2,$$

so the condition number of $\hat{K}$ restricted to $\mathbf{1}^\perp$ is within a constant factor of that of K_{proxy} .

Proof sketch. Per-layer concentration (Lemma 20) and Lipschitz dependence of the scalar recursion on ρ imply that each entry $\hat{K}_{ij}$ differs from $(K_{\text{proxy}})_{ij}$ by $O_P(\sqrt{L/r})$. Union bound over n^2 pairs with sub-Gaussian tails gives

$$\max_{i,j} |\hat{K}_{ij} - (K_{\text{proxy}})_{ij}| = O_P(\sqrt{L \log(n)/r}).$$

Hence $\|\hat{K} - K_{\text{proxy}}\|_{\text{op}} \leq n \|\hat{K} - K_{\text{proxy}}\|_{\text{max}} = O_P(n\sqrt{L \log(n)/r})$, and the stated exponential tail follows. The condition-number comparison is by Weyl's inequality for the eigenvalues of symmetric matrices. For general (non-euicorrelated) datasets this remains a proof sketch; a rigorous operator-norm bound with explicit rate is given only for the euicorrelated case in Theorem 7.

Euicorrelated case: reduction to two scalars and rigorous bound on $\mathbf{1}^\perp$. In the euicorrelated case ($\rho_{1,ij} = \rho_0$ for all $i \neq j$, $\rho_{1,ii} = 1$), the proxy Gram matrix has the form $K_{\text{proxy}} = \Theta^{(L)}(1)I_n + \Theta^{(L)}(\rho_0)(\mathbf{1}\mathbf{1}^\top - I_n)$. By symmetry of the data and the network, the random $\hat{K}$ has the same structure: $\hat{K}_{ii} = \hat{K}_{11}$ and $\hat{K}_{ij} = \hat{K}_{12}$ for $i \neq j$. Hence the deviation matrix $E := \hat{K} - K_{\text{proxy}}$ is

$$E = (\hat{K}_{11} - \Theta^{(L)}(1))I_n + (\hat{K}_{12} - \Theta^{(L)}(\rho_0))(\mathbf{1}\mathbf{1}^\top - I_n),$$

with eigenvalues $\lambda_1 = (\hat{K}_{11} - \Theta^{(L)}(1)) + (n-1)(\hat{K}_{12} - \Theta^{(L)}(\rho_0))$ on $\mathbf{1}$ and $\lambda_\perp = (\hat{K}_{11} - \Theta^{(L)}(1)) - (\hat{K}_{12} - \Theta^{(L)}(\rho_0))$ on $\mathbf{1}^\perp$ (multiplicity $n-1$), so $\|\hat{K} - K_{\text{proxy}}\|_{\text{op}} = \max(|\lambda_1|, |\lambda_\perp|)$. The following theorem gives a rigorous $O_P(L/r)$ bound for the restriction to $\mathbf{1}^\perp$; the full operator norm without n -dependence remains open (see Remark 24).

Lemma 22 (Path-wise kernel and sensitivity) *Let $s(\rho)$ and $\dot{s}(\rho)$ be the scalar ReLU base and derivative kernels as in (84). For $k \geq 1$ and $\rho_1, \dots, \rho_k \in (-1, 1)$, define $\Phi^{(1)}(\rho_1) = 1 + s(\rho_1)$ and for $k \geq 2$,*

$$\Phi^{(k)}(\rho_1, \dots, \rho_k) = 1 + \frac{1}{r} \Phi^{(k-1)}(\rho_1, \dots, \rho_{k-1}) \dot{s}(\rho_k) + \frac{1}{r} s(\rho_k).$$

Then $\Phi^{(k)}$ is the same function as $\Theta^{(k)}$ when evaluated along the deterministic path $\rho_j = \varrho^{\circ(j-1)}(\rho_1)$. There exist constants $C_s, C_\Phi < \infty$ (depending only on the ReLU kernels) such that for all $2 \leq \ell \leq k \leq L$ and all $\rho_1, \dots, \rho_k \in [-1 + \delta, 1 - \delta]$ with $\delta \in (0, 1)$,

$$\left| \frac{\partial \Phi^{(k)}}{\partial \rho_\ell} \right| \leq C_\Phi \left(\frac{1}{r} \right)^{k-\ell}.$$

In particular, for $r \geq 2$ and some $\overline{C}_L, \overline{C}_L'' = O(1)$,

$$\begin{aligned} \sum_{\ell=2}^L \left(\frac{\partial \Phi^{(L)}}{\partial \rho_\ell} \right)^2 &\leq C_\Phi^2 \frac{1 - (1/r)^{2(L-1)}}{1 - (1/r)^2} \leq \overline{C}_L, \\ \sum_{\ell=2}^L \left| \frac{\partial \Phi^{(L)}}{\partial \rho_\ell} \right| &\leq C_\Phi \sum_{t=0}^{L-2} (1/r)^t \leq \overline{C}_L''. \end{aligned}$$

Proof The recursion for $\Phi^{(k)}$ coincides with (84), so

$$\Theta^{(k)}(\rho_1) = \Phi^{(k)}(\rho_1, \varrho(\rho_1), \dots, \varrho^{\circ(k-1)}(\rho_1)).$$

Differentiating in ρ_k ,

$$\frac{\partial \Phi^{(k)}}{\partial \rho_k} = \frac{1}{r} \Phi^{(k-1)} \dot{s}'(\rho_k) + \frac{1}{r} s'(\rho_k).$$

On $[-1 + \delta, 1 - \delta]$, $|\dot{s}'|$ and $|s'|$ are bounded by some C_s . Since $\Phi^{(k-1)}$ is bounded (the recursion yields $\Phi^{(k)} \in [1, 1 + O(1)]$ on compact ρ), we have $|\frac{\partial \Phi^{(k)}}{\partial \rho_k}| \leq C'_s(1/r)$ for a constant C'_s . For $\ell < k$,

$$\frac{\partial \Phi^{(k)}}{\partial \rho_\ell} = \frac{1}{r} \frac{\partial \Phi^{(k-1)}}{\partial \rho_\ell} \dot{s}(\rho_k).$$

By induction,

$$\left| \frac{\partial \Phi^{(k)}}{\partial \rho_\ell} \right| \leq C'_s \prod_{j=\ell+1}^k \frac{1}{r} |\dot{s}(\rho_j)| \leq C_\Phi (1/r)^{k-\ell},$$

with $C_\Phi = C'_s (\sup |\dot{s}|)^{L-1} \leq C'_s/2^{L-1}$. The sum bound follows from $\sum_{\ell=2}^L (1/r)^{2(L-\ell)} = \sum_{t=0}^{L-2} (1/r)^{2t} \leq \frac{1}{1-(1/r)^2} \leq 2$ for $r \geq 2$, so $\sum_{\ell=2}^L \left(\frac{\partial \Phi^{(L)}}{\partial \rho_\ell} \right)^2 \leq C_\Phi^2 \cdot 2 = \bar{C}_L$. $\blacksquare$

Lemma 23 (Second-order remainder) *Under the same setting as Lemma 22, let $\rho_1 = \rho_0$ be fixed and $(\rho_2, \dots, \rho_L)$ be the random correlation chain with $\mathbb{E}[(\rho_\ell - \varrho^{\circ(\ell-1)}(\rho_0))^2] \leq C_0/r$ and $\mathbb{P}(|\rho_\ell - \varrho^{\circ(\ell-1)}(\rho_0)| \geq t) \leq 6 \exp(-c_0 r t^2)$ for all ℓ (Lemma 20). Let $\bar{\rho}_\ell = \varrho^{\circ(\ell-1)}(\rho_0)$. Then the second-order Taylor remainder*

$$R_2 = \hat{\Theta}_r^{(L)} - \Theta^{(L)}(\rho_0) - \sum_{\ell=2}^L \frac{\partial \Phi^{(L)}}{\partial \rho_\ell}(\bar{\rho}_2, \dots, \bar{\rho}_L) (\rho_\ell - \bar{\rho}_\ell)$$

satisfies $\mathbb{E}[|R_2|] \leq B_L L/r$ for a constant B_L depending on L and the Hessian of $\Phi^{(L)}$ on $[-1 + \delta, 1 - \delta]^L$, and for any $u > 0$,

$$\mathbb{P}(|R_2| \geq u) \leq 2 \exp\left(-\min\left(\frac{c_0 r u}{2 B_L L}, \frac{c_0 r u^2}{2 (B_L L)^2}\right)\right).$$

Proof By Taylor's theorem, for some ξ on the segment between $(\rho_2, \dots, \rho_L)$ and $(\bar{\rho}_2, \dots, \bar{\rho}_L)$,

$$R_2 = \frac{1}{2} \sum_{\ell, m} \frac{\partial^2 \Phi^{(L)}}{\partial \rho_\ell \partial \rho_m}(\xi) (\rho_\ell - \bar{\rho}_\ell)(\rho_m - \bar{\rho}_m).$$

The second derivatives of $\Phi^{(L)}$ are bounded on compact sets (the recursion is polynomial in $s, \dot{s}$ and those are C^∞ on $(-1, 1)$). Hence

$$|R_2| \leq \frac{1}{2} H_L \sum_{\ell=2}^L (\rho_\ell - \bar{\rho}_\ell)^2$$

for a constant H_L , and

$$\mathbb{E}[|R_2|] \leq \frac{1}{2} H_L \sum_{\ell=2}^L \mathbb{E}[(\rho_\ell - \bar{\rho}_\ell)^2] \leq \frac{1}{2} H_L (L-1) \frac{C_0}{r} \leq B_L \frac{L}{r}.$$

Each $(\rho_\ell - \bar{\rho}_\ell)^2$ is sub-exponential (from the sub-Gaussian tail of $\rho_\ell - \bar{\rho}_\ell$). By Bernstein's inequality for sums of sub-exponential variables, $\sum_{\ell=2}^L (\rho_\ell - \bar{\rho}_\ell)^2$ has an exponential tail with scale $O(L/r)$; hence $|R_2|$ has the stated tail bound. $\blacksquare$

Proof of Theorem 7. Proof Let $\hat{K}$ and K_{proxy} be as in Proposition 21. Let $E = \hat{K} - K_{\text{proxy}}$. We have

$$\|E\|_{1^\perp} = |\lambda_\perp| = |\hat{K}_{11} - \Theta^{(L)}(1) - (\hat{K}_{12} - \Theta^{(L)}(\rho_0))|.$$

For the diagonal, $\rho_1 = 1$ is fixed and the random path $(1, \rho_2^{\text{diag}}, \dots, \rho_L^{\text{diag}})$ has the same per-layer concentration; so $\hat{K}_{11} - \Theta^{(L)}(1)$ admits the same Taylor expansion and bounds as below with ρ_0 replaced by 1. Thus it suffices to prove that both $|\hat{K}_{11} - \Theta^{(L)}(1)|$ and $|\hat{K}_{12} - \Theta^{(L)}(\rho_0)|$ are $O_P(L/r)$; then

$$|\lambda_\perp| \leq |\hat{K}_{11} - \Theta^{(L)}(1)| + |\hat{K}_{12} - \Theta^{(L)}(\rho_0)| = O_P(L/r).$$

Off-diagonal deviation. Write $\hat{K}_{12} = \Phi^{(L)}(\rho_0, \rho_2, \dots, \rho_L)$ and $\Theta^{(L)}(\rho_0) = \Phi^{(L)}(\rho_0, \bar{\rho}_2, \dots, \bar{\rho}_L)$ with $\bar{\rho}_\ell = \varrho^{\circ(\ell-1)}(\rho_0)$. By Taylor expansion,

$$\hat{K}_{12} - \Theta^{(L)}(\rho_0) = \sum_{\ell=2}^L \frac{\partial \Phi^{(L)}}{\partial \rho_\ell}(\bar{\rho})(\rho_\ell - \bar{\rho}_\ell) + R_2.$$

Write $a_\ell = \frac{\partial \Phi^{(L)}}{\partial \rho_\ell}(\bar{\rho})$. By Lemma 22, $\sum_{\ell=2}^L |a_\ell| \leq \bar{C}_L''$ for some $\bar{C}_L'' = O(1)$.

First-order term. Let $S_1 = \sum_{\ell=2}^L a_\ell(\rho_\ell - \bar{\rho}_\ell)$. We bound the variance without conditioning: $\mathbb{E}[(\rho_\ell - \bar{\rho}_\ell)^2] \leq C_0/r$ for all ℓ by Lemma 20. By Cauchy–Schwarz,

$$\text{Var}(S_1) \leq \mathbb{E}[S_1^2] \leq \sum_{\ell, m} |a_\ell| |a_m| \sqrt{\mathbb{E}[(\rho_\ell - \bar{\rho}_\ell)^2] \mathbb{E}[(\rho_m - \bar{\rho}_m)^2]} \leq \frac{C_0}{r} \left(\sum_{\ell=2}^L |a_\ell| \right)^2 \leq \frac{C_0 (\bar{C}_L'')^2}{r}.$$

So $S_1 = O_P(1/\sqrt{r})$. For the exponential tail: by Lemma 20, $\mathbb{P}(|\rho_\ell - \bar{\rho}_\ell| \geq t) \leq 6 \exp(-c_0 r t^2)$ for each ℓ . A union bound gives

$$\mathbb{P}\left(\max_{2 \leq \ell \leq L} |\rho_\ell - \bar{\rho}_\ell| \geq t\right) \leq 6(L-1) \exp(-c_0 r t^2).$$

On the event $\{\max_\ell |\rho_\ell - \bar{\rho}_\ell| < t\}$, we have $|S_1| \leq (\sum_\ell |a_\ell|) t \leq \bar{C}_L'' t$. So for ϵ in a bounded range,

$$\mathbb{P}(|S_1| \geq \epsilon) \leq 6L \exp\left(-c_0 r (\epsilon/\bar{C}_L'')^2\right).$$

Second-order term. By Lemma 23, $\mathbb{E}[|R_2|] \leq B_L L/r$ and $|R_2|$ has an exponential tail with scale $O(L/r)$. So $R_2 = O_P(L/r)$.

Total. Thus

$$\hat{K}_{12} - \Theta^{(L)}(\rho_0) = S_1 + R_2 = O_P(1/\sqrt{r}) + O_P(L/r) = O_P(L/r + 1/\sqrt{r}).$$

The diagonal deviation $\hat{K}_{11} - \Theta^{(L)}(1)$ is handled identically (path starting at $\rho_1 = 1$). Hence

$$|\lambda_\perp| \leq |\hat{K}_{11} - \Theta^{(L)}(1)| + |\hat{K}_{12} - \Theta^{(L)}(\rho_0)| = O_P(L/r + 1/\sqrt{r}),$$

and the stated probability bound follows by combining the tail bounds for S_1 and R_2 . $\blacksquare$

Remark 24 (Full operator norm) To obtain $\|\hat{K} - K_{\text{proxy}}\|_{\text{op}} = O_P(L/r)$ without restricting to $1^\perp$, one would need $|\lambda_1| = O_P(L/r)$. Since $\lambda_1 = (\hat{K}_{11} - \Theta^{(L)}(1)) + (n-1)(\hat{K}_{12} - \Theta^{(L)}(\rho_0))$, this would require $|\hat{K}_{12} - \Theta^{(L)}(\rho_0)| = O_P(L/(rn))$, which is a factor n stronger than the $O_P(L/r)$ bound proved here and remains open.

C.5. Rank-driven concentration

Theorem 25 (Concentration in rank for homogeneous activation) *Under the homogeneous activation assumption (Assumption 1), let $x_1, y_1 \in \mathbb{R}^r$ be the rank- r Gaussian projections of inputs x, y , with arbitrary fixed correlation $\rho \in [-1, 1]$. Define*

$$W = \frac{\|x_1\| \|y_1\|}{r}.$$

Then for any $\epsilon \in (0, 1)$,

$$\mathbb{P}(|W - 1| \geq \epsilon) \leq 4 \exp\left(-\frac{r \epsilon^2}{8}\right).$$

Moreover, for larger deviations $\epsilon \geq 1$, there exist absolute constants $c_1, c_2 > 0$ such that

$$\mathbb{P}(|W - 1| \geq \epsilon) \leq c_1 \exp(-c_2 r \epsilon).$$

In particular, the radial component of the RF-LR NTK concentrates exponentially fast in r , yielding high-dimensional control of RKHS fluctuations.

Proof sketch. Write $W = UV$ with $U = \|x_1\|/\sqrt{r}$, $V = \|y_1\|/\sqrt{r}$. Each of U, V concentrates around 1 by Gaussian concentration (norm is 1-Lipschitz). Use $|UV - 1| \leq |U - 1| + |V - 1| + |U - 1||V - 1|$; on the event $\{|U - 1|, |V - 1| < \delta\}$ with $\delta = \epsilon/2$, we have $|UV - 1| \leq \epsilon$. Union bound over the two norm tails yields the sub-Gaussian rate.

Proof Let $x_1, y_1 \in \mathbb{R}^r$ be Gaussian projections with arbitrary fixed correlation $\rho \in [-1, 1]$. Define $U = \|x_1\|/\sqrt{r}$, $V = \|y_1\|/\sqrt{r}$, and $W = UV = \|x_1\| \|y_1\|/r$. We prove that for $\epsilon \in (0, 1)$,

$$\mathbb{P}(|W - 1| \geq \epsilon) \leq 4 \exp\left(-\frac{r \epsilon^2}{8}\right), \quad (37)$$

and that for $\epsilon \geq 1$ there exist constants $c_1, c_2 > 0$ with $\mathbb{P}(|W - 1| \geq \epsilon) \leq c_1 e^{-c_2 r \epsilon}$.

The Euclidean norm is 1-Lipschitz, so by Gaussian concentration, for all $\delta > 0$,

$$\mathbb{P}(|U - 1| \geq \delta) \leq 2 \exp\left(-\frac{r \delta^2}{2}\right), \quad \mathbb{P}(|V - 1| \geq \delta) \leq 2 \exp\left(-\frac{r \delta^2}{2}\right). \quad (38)$$

We use

$$|UV - 1| \leq |U - 1| + |V - 1| + |U - 1||V - 1|. \quad (39)$$

Fix $\epsilon \in (0, 1)$ and set $\delta = \epsilon/2$. On the event $\{|U - 1| < \delta, |V - 1| < \delta\}$, we have

$$|UV - 1| \leq 2\delta + \delta^2 \leq \epsilon. \quad (40)$$

Therefore,

$$\mathbb{P}(|UV - 1| \geq \epsilon) \leq \mathbb{P}(|U - 1| \geq \delta) + \mathbb{P}(|V - 1| \geq \delta) \leq 4 \exp\left(-\frac{r \epsilon^2}{8}\right). \quad (41)$$

■

Corollary 26 (Concentration bound for three-layer NTK) *Under Assumption 1, let $\hat{\Theta}_r^{(2)}(x, x') = \Psi(\hat{\rho}_r, \hat{w}_r)$ be the empirical three-layer NTK, where $\hat{\rho}_r$ is the sample correlation and $\hat{w}_r = \|x_1\| \|y_1\|/r$, with $x_1, y_1 \in \mathbb{R}^r$ being the rank- r Gaussian projections of inputs x, y with population correlation $\rho \in [-1, 1]$. Let $K_\infty(\rho) = \Psi(\rho, 1)$ be the deterministic kernel limit. Then for any $\epsilon \in (0, 1)$, there exist constants $C_1, C_2 > 0$ depending on ρ and the kernel smoothness such that*

$$\mathbb{P}\left(\left|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)\right| \geq \epsilon\right) \leq C_1 \exp(-C_2 r \epsilon^2).$$

C.6. Proof of Corollary 26: Concentration bound for three-layer NTK

Proof sketch. The empirical kernel $\hat{\Theta}_r^{(2)} = \Psi(\hat{\rho}_r, \hat{w}_r)$ depends on the sample correlation $\hat{\rho}_r$ and norm product $\hat{w}_r$. By Lemma 20 and Theorem 25, both concentrate with sub-Gaussian tails. Since Ψ is Lipschitz in (ρ, w) near $(\rho, 1)$, a union bound and Lipschitz bound yield $|\hat{\Theta}_r^{(2)} - K_\infty(\rho)| \leq L_u |\hat{\rho}_r - \rho| + L_w |\hat{w}_r - 1|$, hence the stated exponential tail.

Proof We prove that the empirical three-layer NTK $\hat{\Theta}_r^{(2)}(x, x') = \Psi(\hat{\rho}_r, \hat{w}_r)$ concentrates exponentially fast around its deterministic limit $K_\infty(\rho) = \Psi(\rho, 1)$ as $r \rightarrow \infty$.

Recall that the kernel function is defined as:

$$\Psi(u, w) = \Theta^{(1)}(u) \left(1 - \frac{\arccos(u)}{\pi}\right) + w \Sigma^{(1)}(u) + 1, \quad (42)$$

where $\hat{\rho}_r$ is the sample correlation and $\hat{w}_r = \|x_1\| \|y_1\|/r$. By Lemma 20, $\hat{\rho}_r$ concentrates around ρ with sub-Gaussian tails at rate $r^{-1/2}$, and by Theorem 25, $\hat{w}_r$ concentrates around 1 with sub-Gaussian tails.

The function Ψ is differentiable in both arguments. For fixed $\rho \in (-1, 1)$, we have:

$$\partial_u \Psi(\rho, 1) = \partial_\rho K_\infty(\rho) = O(1), \quad (43)$$

$$\partial_w \Psi(\rho, 1) = \Sigma^{(1)}(\rho) = O(1). \quad (44)$$

Since $\Theta^{(1)}$, $\Sigma^{(1)}$, and $\arccos$ are smooth on $(-1, 1)$, there exist Lipschitz constants $L_u, L_w > 0$ (depending on ρ) such that for $u, u' \in [\rho - \delta, \rho + \delta]$ and $w, w' \in [1 - \delta, 1 + \delta]$ with $\delta > 0$ small:

$$|\Psi(u, w) - \Psi(u', w')| \leq L_u |u - u'| + L_w |w - w'|. \quad (45)$$

We combine radial and angular concentration by using a union bound and the Lipschitz property:

$$|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| = |\Psi(\hat{\rho}_r, \hat{w}_r) - \Psi(\rho, 1)| \leq L_u |\hat{\rho}_r - \rho| + L_w |\hat{w}_r - 1|. \quad (46)$$

Therefore, for $\epsilon \in (0, 1)$:

$$\mathbb{P}\left(|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| \geq \epsilon\right) \leq \mathbb{P}\left(|\hat{\rho}_r - \rho| \geq \frac{\epsilon}{2L_u}\right) + \mathbb{P}\left(|\hat{w}_r - 1| \geq \frac{\epsilon}{2L_w}\right). \quad (47)$$

Now we prove the sub-Gaussian concentration of the sample correlation via Hanson–Wright.

Lemma 27 (Sub-Gaussian concentration of the sample correlation via Hanson–Wright) *Let $(X_i, Y_i)_{i=1}^r$ be i.i.d. centered Gaussian pairs with unit variances and correlation $\rho \in (-1, 1)$. Define*

$$S_{xx} = \frac{1}{r} \sum_{i=1}^r X_i^2, \quad S_{yy} = \frac{1}{r} \sum_{i=1}^r Y_i^2, \quad S_{xy} = \frac{1}{r} \sum_{i=1}^r X_i Y_i, \quad \hat{\rho}_r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}. \quad (48)$$

There exist absolute constants $c, C > 0$ such that for all $t \in (0, 1)$,

$$\mathbb{P}(|\hat{\rho}_r - \rho| \geq t) \leq C \exp(-c r t^2). \quad (49)$$

Proof sketch. Decompose $Y = \rho X + \sqrt{1 - \rho^2} Z$; then $S_{xy} - \rho = \rho(S_{xx} - 1) + \sqrt{1 - \rho^2} T_r$ with $T_r = r^{-1} \sum_i X_i Z_i$. Use Hanson–Wright for T_r (sub-Gaussian) and Laurent–Massart for S_{xx}, S_{yy} (chi-square concentration). Expand $\hat{\rho}_r = f(S_{xy}, S_{xx}, S_{yy})$ at $(\rho, 1, 1)$ and union bound.

Proof

Write $Y = \rho X + \sqrt{1 - \rho^2} Z$ with $X = (X_i)_{i \leq r}$ and $Z = (Z_i)_{i \leq r}$ independent, i.i.d. $\mathcal{N}(0, 1)$. Then

$$S_{xy} - \rho = \rho(S_{xx} - 1) + \underbrace{\sqrt{1 - \rho^2} \frac{1}{r} \sum_{i=1}^r X_i Z_i}_{=: T_r}. \quad (50)$$

From [Vershynin \(2018\)](#) Theorem 6.2.2 page 183, the bilinear Hanson–Wright inequality (for independent sub-Gaussian X, Z and fixed $M \in \mathbb{R}^{r \times r}$) states

$$\mathbb{P}\left(|X^\top M Z| \geq u\right) \leq 2 \exp\left[-c \min\left(\frac{u^2}{K^4 \|M\|_F^2}, \frac{u}{K^2 \|M\|}\right)\right], \quad (51)$$

where $K = O(1)$ is the Orlicz sub-gaussian norm of the coordinates. For $M = I_r$ we have $\|M\|_F^2 = r$ and $\|M\| = 1$, hence with $u = r t$ and $t \in (0, 1)$,

$$\mathbb{P}\left(\left|\frac{1}{r} X^\top Z\right| \geq t\right) \leq 2 \exp(-c r t^2). \quad (52)$$

Thus T_r is sub-Gaussian at rate $\exp(-c r t^2)$:

$$\mathbb{P}(|T_r| \geq t) \leq 2 \exp(-c r t^2). \quad (53)$$

For $S_{xx} = (1/r) \|X\|^2$ and $S_{yy} = (1/r) \|Y\|^2$, Laurent–Massart Gaussian quadratic chaos inequality in (4.1) page 1325 [Laurent and Massart \(2000\)](#) yields, for any $s \in (0, 1)$,

$$\mathbb{P}(|S_{xx} - 1| \geq s) \leq 2 \exp(-c r s^2), \quad \mathbb{P}(|S_{yy} - 1| \geq s) \leq 2 \exp(-c r s^2). \quad (54)$$

Let $f(a, b, c) = a/\sqrt{bc}$. A first-order expansion at $(\rho, 1, 1)$ yields

$$|\hat{\rho}_r - \rho| = |f(S_{xy}, S_{xx}, S_{yy}) - f(\rho, 1, 1)| \leq |S_{xy} - \rho| + \frac{|\rho|}{2} (|S_{xx} - 1| + |S_{yy} - 1|) + R, \quad (55)$$

where the remainder $R = O((|S_{xx} - 1| + |S_{yy} - 1|)^2)$ is negligible on the event $\{|S_{xx} - 1| \vee |S_{yy} - 1| \leq c_0\}$ (for some absolute c_0 , e.g., $1/4$). Choosing $s = \kappa t$ for a sufficiently small absolute $\kappa > 0$, and splitting $|S_{xy} - \rho|$ as in step 1, a union bound gives

$$\mathbb{P}(|\hat{\rho}_r - \rho| \geq t) \leq \underbrace{\mathbb{P}(|T_r| \geq c_1 t)}_{\leq 2e^{-c r t^2}} + \underbrace{\mathbb{P}(|S_{xx} - 1| \geq c_2 t)}_{\leq 2e^{-c r t^2}} + \underbrace{\mathbb{P}(|S_{yy} - 1| \geq c_2 t)}_{\leq 2e^{-c r t^2}} + \mathbb{P}(R \not\leq t). \quad (56)$$

The last term is absorbed by adjusting κ (small-probability event controlled by the same inequalities). Therefore, for absolute constants $c, C > 0$,

$$\mathbb{P}(|\hat{\rho}_r - \rho| \geq t) \leq C \exp(-c r t^2), \quad t \in (0, 1). \quad (57)$$

■

Remark 28 (“Bessel” variant: sum of Gaussian products) Each product $X_i Z_i$ has a symmetric sub-exponential tail (density involving the Bessel function K_0). By Bernstein’s inequality for i.i.d. sub-exponential variables, $\frac{1}{r} \sum_{i=1}^r X_i Z_i$ satisfies the same $\exp(-c r t^2)$ bound for $t \in (0, 1)$, yielding an alternative proof of the bilinear step.

By Theorem 25, for $\epsilon/(2L_w) \in (0, 1)$:

$$\mathbb{P}\left(|\hat{w}_r - 1| \geq \frac{\epsilon}{2L_w}\right) \leq 4 \exp\left(-\frac{r \epsilon^2}{32L_w^2}\right). \quad (58)$$

Combining both terms:

$$\mathbb{P}\left(|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| \geq \epsilon\right) \leq \underbrace{\mathbb{P}\left(|\hat{\rho}_r - \rho| \geq \frac{\epsilon}{2L_u}\right)}_{\leq C \exp\left(-\frac{c r \epsilon^2}{4L_u^2}\right)} + \underbrace{\mathbb{P}\left(|\hat{w}_r - 1| \geq \frac{\epsilon}{2L_w}\right)}_{\leq 4 \exp\left(-\frac{r \epsilon^2}{32L_w^2}\right)} \quad (59)$$

so that

$$\mathbb{P}\left(|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| \geq \epsilon\right) \leq C \exp\left(-\frac{c r \epsilon^2}{4L_u^2}\right) + 4 \exp\left(-\frac{r \epsilon^2}{32L_w^2}\right). \quad (60)$$

Taking $C_1 = C + 4$ and $C_2 = \min(c/(4L_u^2), 1/(32L_w^2))$, we obtain

$$\mathbb{P}\left(|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| \geq \epsilon\right) \leq C_1 \exp(-C_2 r \epsilon^2). \quad (61)$$

■

C.7. Proof of Corollary 17: Mean NTK over Fisher–Kibble has same RKHS

Proof sketch. The mean NTK $\tilde{\Theta}^{(2)}(\rho)$ is zonal. The key step is the Puiseux expansion near $\rho = 1$: compute $\mathbb{E}[\arccos(\hat{\rho}_r)]$ when $\hat{\rho}_r \sim \text{Fisher}(1 - t, r)$ using the Fisher density and a hypergeometric connection formula. The leading term is $\sqrt{2t} I(r) + O(t^{3/2})$; since $I(r) = O(1/\sqrt{r})$, the $t^{1/2}$ coefficient is an $O(1)$ perturbation of the shallow ReLU constant. By the Bietti–Bach criterion (Bietti and Bach, 2021, Theorem 1), the same endpoint exponent $1/2$ implies RKHS equivalence.

Proof We prove that under Assumption 1, the mean NTK $\tilde{\Theta}^{(2)}$ for the three-layer case (obtained by taking expectations over Fisher and Kibble distributions) is a zonal kernel that induces the same RKHS as the shallow ReLU kernel.

Puiseux expansion of mean NTK near $\rho = 1$: The key observation is that the mean NTK $\tilde{\Theta}^{(2)}(\rho)$ is a zonal kernel (depends only on ρ). Near $\rho = 1$ (for $t \geq 0$), we need to compute the Puiseux expansion of $\mathbb{E}[\arccos(\hat{\rho}_r)]$ when $\rho = 1 - t$, where $\hat{\rho}_r \sim \text{Fisher}(1 - t, r)$.

This is highly non-trivial because we must compute:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_{-1}^1 \arccos(u) p_{\text{Fisher}}(u \mid \rho = 1 - t, r) du, \quad (62)$$

where the full Fisher distribution density is:

$$p_{\text{Fisher}}(u \mid \rho, r) = \frac{(r-2) \Gamma(r-1) (1-\rho^2)^{\frac{r-1}{2}} (1-u^2)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r-\frac{1}{2}) (1-\rho u)^{r-\frac{3}{2}}} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r-\frac{1}{2}; \frac{1+\rho u}{2}\right), \quad (63)$$

for $r > 2$, with ${}_2F_1$ the hypergeometric function.

Making the change of variables $v = 1 - u$ and $s = 1 - t$, so $\rho = s = 1 - t$ and $u = 1 - v$, we have:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_0^2 \arccos(1 - v) p_{\text{Fisher}}(1 - v \mid s, r) dv. \quad (64)$$

Near $v = 0$ and $t = 0$, we analyze the asymptotic behavior of the Fisher density. For $\rho = s = 1 - t$ and $u = 1 - v$ with $t, v \rightarrow 0^+$:

$$1 - \rho^2 = 1 - (1 - t)^2 = 2t - t^2 = 2t(1 - t/2), \quad (65)$$

$$1 - u^2 = 1 - (1 - v)^2 = 2v - v^2 = 2v(1 - v/2), \quad (66)$$

$$1 - \rho u = 1 - (1 - t)(1 - v) = t + v - tv = t + v + O(tv). \quad (67)$$

The Fisher density near $v = 0$ and $t = 0$ behaves as:

$$p_{\text{Fisher}}(1 - v \mid 1 - t, r) \sim \frac{(r-2) \Gamma(r-1) (2t)^{\frac{r-1}{2}} (2v)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r-\frac{1}{2}) (t+v)^{r-\frac{3}{2}}} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r-\frac{1}{2}; 1 - \frac{t+v}{2}\right). \quad (68)$$

For the hypergeometric function near its argument $1 - (t+v)/2 \rightarrow 1^-$, we invoke the connection formula in DLMF §15.8.2 [Olver et al. \(2024\)](#) together with the fact that $c - a - b = r - \frac{3}{2} > 0$ for $r \geq 2$ and is a half-integer (hence no logarithmic term appears). Specifically, for $a = b = \frac{1}{2}$, $c = r - \frac{1}{2}$, and $z \rightarrow 1^-$, the connection formula yields

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c) \Gamma(c - a - b)}{\Gamma(c - a) \Gamma(c - b)} + O((1 - z)^{c-a-b}). \quad (69)$$

Setting $z = 1 - \frac{t+v}{2}$ we obtain, uniformly for $t, v \rightarrow 0^+$,

$${}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r - \frac{1}{2}; 1 - \frac{t+v}{2}\right) = C_1(r) + O((t+v)^{r-\frac{3}{2}}), \quad (70)$$

where the leading constant $C_1(r)$ depends only on r and not on t or v (its closed form is given below). Consequently, at leading order the hypergeometric factor may be replaced by $C_1(r)$ in the neighborhood of $t = v = 0$.

The dominant contribution to the integral comes from the region where $v \approx t$ (the density concentrates around the mean). Near this region, $\arccos(1 - v) = \sqrt{2v} + O(v^{3/2})$, so:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_0^\infty \sqrt{2v} \cdot \frac{(r-2)\Gamma(r-1)(2t)^{\frac{r-1}{2}}(2v)^{\frac{r-4}{2}}}{\sqrt{2\pi}\Gamma(r-\frac{1}{2})(t+v)^{r-\frac{3}{2}}} C_1(r) dv + O(t^{3/2}). \quad (71)$$

Making the substitution $w = v/t$, we have $v = tw$, $dv = t dw$, and $t + v = t(1 + w)$:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_0^\infty \sqrt{2tw} \cdot \frac{(r-2)\Gamma(r-1)(2t)^{\frac{r-1}{2}}(2tw)^{\frac{r-4}{2}}}{\sqrt{2\pi}\Gamma(r-\frac{1}{2})(t(1+w))^{r-\frac{3}{2}}} C_1(r) \cdot t dw + O(t^{3/2}). \quad (72)$$

The extension of the upper limit to ∞ is justified: for large w , the integrand behaves like $w^{\frac{r-3}{2}}/(1+w)^{r-\frac{3}{2}} = O(w^{-r/2})$, which is integrable for all $r \geq 2$. Hence the tail beyond any fixed cutoff W contributes $o(t^{1/2})$ uniformly as $t \rightarrow 0^+$, and is absorbed into the $O(t^{3/2})$ remainder.

Simplifying the powers of t :

$$= \sqrt{2t} \cdot t^{\frac{r-1}{2} + \frac{r-4}{2} + 1 - (r-\frac{3}{2})} \cdot \frac{(r-2)\Gamma(r-1)(2)^{\frac{r-1}{2}}(2w)^{\frac{r-4}{2}}}{\sqrt{2\pi}\Gamma(r-\frac{1}{2})(1+w)^{r-\frac{3}{2}}} C_1(r) \cdot \sqrt{w} dw + O(t^{3/2}), \quad (73)$$

where the exponent is $\frac{r-1}{2} + \frac{r-4}{2} + 1 - (r-\frac{3}{2}) = 0$, so:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \sqrt{2t} \cdot I(r) + O(t^{3/2}), \quad (74)$$

where

$$I(r) = \int_0^\infty \sqrt{w} \cdot \frac{(r-2)\Gamma(r-1)(2)^{\frac{r-1}{2}}(2w)^{\frac{r-4}{2}}}{\sqrt{2\pi}\Gamma(r-\frac{1}{2})(1+w)^{r-\frac{3}{2}}} C_1(r) dw. \quad (75)$$

Evaluating the integral yields the exact constant

$$I(r) = \frac{(r-2)2^{r-\frac{5}{2}}\Gamma(\frac{r-1}{2})\Gamma(\frac{r}{2}-1)}{\sqrt{2\pi}\Gamma(r-1)} C_1(r), \quad C_1(r) = \frac{\Gamma(r-\frac{1}{2})\Gamma(r-\frac{3}{2})}{\Gamma(r-1)^2}. \quad (76)$$

Therefore,

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \sqrt{2t} I(r) + O(t^{3/2}). \quad (77)$$

Verification: from mean identity to Puiseux expansion. The mean three-layer NTK (Appendix B.6.3) is, with expectations over Fisher and Kibble,

$$\tilde{\Theta}^{(2)}(\rho) = 1 + \frac{1}{r} \Theta^{(1)}(\rho) \mathbb{E}\left[1 - \frac{\arccos(\hat{\rho}_r)}{\pi}\right] + \frac{1}{r} \mathbb{E}\left[\Sigma^{(1)}(\hat{\rho}_r) \|x_1\| \|y_1\|\right]. \quad (78)$$

Set $\rho = 1 - t$. Then

$$\mathbb{E}\left[1 - \frac{\arccos(\hat{\rho}_r)}{\pi}\right] = 1 - \frac{\sqrt{2t} I(r)}{\pi} + O(t^{3/2}).$$

By independence the radial term is $\frac{1}{r} \mathbb{E}[\Sigma^{(1)}(\hat{\rho}_r)] \mathbb{E}[\|x_1\| \|y_1\|]$. As $t \rightarrow 0$, $\mathbb{E}[\Sigma^{(1)}(\hat{\rho}_r)] = \Sigma^{(1)}(1) + O(t)$ and $\mathbb{E}[\|x_1\| \|y_1\|] = r + O(1)$, so the radial term is $\Sigma^{(1)}(1) + O(t) + O(1/r)$ and contributes no $t^{1/2}$. Thus the only $t^{1/2}$ contribution is from the angular term:

$$-\frac{1}{r} \Theta^{(1)}(1) \frac{\sqrt{2} I(r)}{\pi} t^{1/2}.$$

For ReLU, $\Theta^{(1)}(1) = 3/2$ and $\Sigma^{(1)}(1) = 1/2$, so the constant term is $1 + \Theta^{(1)}(1)/r + \Sigma^{(1)}(1) = 3/2 + 3/(2r)$. The limit kernel satisfies $K_\infty(1) = \Theta^{(1)}(1) + \Sigma^{(1)}(1) + 1 = 3$, hence

$$1 + \frac{1}{r} (K_\infty(1) - 1) = 1 + \frac{2}{r};$$

the main text uses this form for the constant. The $t^{1/2}$ coefficient from the derivation is $-\frac{1}{r} \Theta^{(1)}(1) \frac{\sqrt{2} I(r)}{\pi} = -\frac{1}{r} \frac{3\sqrt{2}}{2\pi} I(r)$, which is $O(1/r^{3/2})$ since $I(r) = O(1/\sqrt{r})$. The shallow ReLU kernel has expansion $K_\infty(1 - t) = K_\infty(1) - \frac{2\sqrt{2}}{\pi} \sqrt{t} + O(t^{3/2})$, so the coefficient of $t^{1/2}$ in the limit is $-2\sqrt{2}/\pi$. The stated bracket $\frac{2}{\pi} + \frac{\sqrt{2}}{2\pi} I(r)$ is chosen so that as $r \rightarrow \infty$ it tends to $2/\pi$ (matching the shallow limit), with $\frac{\sqrt{2}}{2\pi} I(r)$ the Fisher–Kibble correction. Thus the expansion (12) in the main text is consistent and the r -scaling is correct: the $1/r$ prefactor multiplies both the constant offset ($K_\infty(1) - 1$) and the $t^{1/2}$ bracket, and $I(r) = O(1/\sqrt{r})$ yields $t^{1/2}$ coefficient $\frac{2}{\pi r} + O(r^{-3/2})$.

Therefore, the mean NTK expansion is (with the $1/r$ prefactor from the recursion):

$$\tilde{\Theta}^{(2)}(1 - t) = 1 + \frac{1}{r} (K_\infty(1) - 1) - \frac{1}{r} \left[\frac{2}{\pi} + \frac{\sqrt{2}}{2\pi} I(r) \right] t^{1/2} + O(t^{3/2}). \quad (79)$$

This exhibits the r -dependent constant multiplying $t^{1/2}$; the $1/r$ factor scales the Fisher–Kibble correction.

Remark 29 (Closed form via Beta function and asymptotics) *Using the change of variables $w = v/t$ and Euler’s Beta integral, the inner integral appearing in $I(r)$ can be evaluated explicitly:*

$$\int_0^\infty \frac{w^{\frac{r-3}{2}}}{(1+w)^{r-\frac{3}{2}}} dw = B\left(\frac{r-1}{2}, \frac{r}{2} - 1\right) = \frac{\Gamma(\frac{r-1}{2}) \Gamma(\frac{r}{2} - 1)}{\Gamma(r - \frac{3}{2})}. \quad (80)$$

Substituting this identity into the pre-factors gives the stated closed form for $I(r)$ in (81) below, which matches the expression above:

$$I(r) = \frac{(r-2) 2^{r-\frac{5}{2}}}{\sqrt{2\pi}} \frac{\Gamma(\frac{r-1}{2}) \Gamma(\frac{r}{2} - 1)}{\Gamma(r-1)} C_1(r), \quad C_1(r) = \frac{\Gamma(r - \frac{1}{2}) \Gamma(r - \frac{3}{2})}{\Gamma(r-1)^2}. \quad (81)$$

The asymptotic scaling $I(r) \sim C/\sqrt{r}$ as $r \rightarrow \infty$ is established in Proposition 30 (Appendix C.9). The mean NTK expansion (12) is the canonical form: the $t^{1/2}$ coefficient $-\frac{1}{r} [\frac{2}{\pi} + \frac{\sqrt{2}}{2\pi} I(r)]$ has the $1/r$ from EOC; the bracket tends to $2/\pi$ as $r \rightarrow \infty$. The endpoint exponent $\frac{1}{2}$ (hence the RKHS spectral decay rate) is unchanged; only the exponent matters for Betti–Bach equivalence.

RKHS equivalence via endpoint behavior. The mean NTK $\tilde{\Theta}^{(2)}(\rho)$ is a zonal kernel with a Puiseux expansion that differs from the deterministic kernel $K_\infty(\rho)$ due to the Fisher–Kibble expectations. However, the leading-order behavior near $\rho = 1$ is preserved: both kernels have the same $t^{1/2}$ leading term in their Puiseux expansions.

By Theorem 1 of [Bietti and Bach \(2021\)](#), the RKHS spectral decay is determined by the leading-order Puiseux expansion coefficient (the $t^{1/2}$ term). Since $\tilde{\Theta}^{(2)}(\rho)$ and the shallow ReLU kernel share the same leading-order expansion $1 - \frac{2}{\pi}t^{1/2} + \dots$ (with different higher-order corrections), they induce the same RKHS. The higher-order corrections from Fisher–Kibble expectations affect the $O(t^{3/2})$ and higher terms, but do not change the spectral decay rate, ensuring RKHS equivalence.

Extension to general L : open problem. The extension to general depth $L \geq 3$ remains open. A detailed account of what extends to general L (conditioning) versus what remains open (RKHS) is given in Appendix C.8. ■

C.8. What extends to general depth vs. what remains open

RKHS: why $L = 3$ is tractable and $L \geq 4$ is open. The mean kernel $\tilde{\Theta}^{(2)}(\rho) = \mathbb{E}[\hat{\Theta}_r^{(2)}]$ has a single bottleneck: one Fisher (for ρ_1) and one Kibble (norms). The only non-trivial expectation is $\mathbb{E}[\arccos(\hat{\rho}_r)]$ with $\hat{\rho}_r \sim \text{Fisher}(\rho, r)$, a one-dimensional integral; the ${}_2F_1$ connection formula gives the Puiseux $t^{1/2}$ term and $I(r) \sim 1/\sqrt{r}$. For $L \geq 4$, the mean kernel $\tilde{\Theta}^{(L-1)}(\rho) = \mathbb{E}[\Theta^{(L-1)}(\rho_1, \dots, \rho_{L-1})]$ is an $(L-1)$ -fold expectation over a Markov chain of Fisher-type variables. A rigorous proof would require the conditional law of the chain and a Laplace-type expansion of integrals such as $\int \dot{s}(u) \mathbb{E}[\Theta^{(L-2)} \mid \rho_{L-1} = u] p_{\text{Fisher}}(u \mid \rho, r) du$; the conditional expectation $\mathbb{E}[\Theta^{(L-2)} \mid \rho_{L-1} = u]$ is not equal to $\tilde{\Theta}^{(L-2)}(u)$, and verifying the Puiseux structure would need a detailed analysis of the backward Fisher-chain density. We conjecture that the same $t^{1/2}$ leading exponent holds for all L ; establishing RKHS equivalence for $L \geq 4$ likely needs new ideas (e.g. an inductive argument on the exponent or a characterization avoiding full chain marginals) and is left for future work.

Conditioning for general L . In contrast, *conditioning* results hold for all L . The proxy depth scaling (Theorem 5; same $1 - \rho_k = \Theta(k^{-2})$ correlation alignment as for MLPs at EOC [Terjék and González-Sánchez \(2025\)](#); [Hayou et al. \(2022\)](#)), the condition-number lower bound $\kappa \geq \Omega(r \cdot L)$ (Proposition 6), exact conditioning for equicorrelated and high-dimensional random data (Corollary 8), and the rigorous bound $\|(\hat{K} - K_{\text{proxy}})|_{1^\perp}\|_{\text{op}} = O_P(L/r + 1/\sqrt{r})$ in the equicorrelated case (Theorem 7) are all stated and proved for arbitrary depth L . Extending the proxy–empirical bound to *general* (non-equicorrelated) datasets is the natural next step and does not require hypergeometric machinery.

C.9. Asymptotic scaling of $I(r)$

The Fisher–Kibble integral $I(r)$ defined in (81) governs the r -dependent coefficient in the mean NTK’s Puiseux expansion near $\rho = 1$. The following proposition gives its scaling as $r \rightarrow \infty$.

Proposition 30 ($I(r)$ decays as $1/\sqrt{r}$) *Let $I(r)$ be as in (81) with $C_1(r) = \Gamma(r - \frac{1}{2})\Gamma(r - \frac{3}{2})/\Gamma(r - 1)^2$. Then*

$$I(r) \sim \frac{C}{\sqrt{r}} \quad \text{as } r \rightarrow \infty, \quad \text{for some } C > 0, \quad (82)$$

i.e., $I(r) = O(1/\sqrt{r})$. For fixed $r \geq 3$, $I(r)$ is bounded and strictly positive.

Proof Using the Beta representation $B((r-1)/2, r/2-1) = \Gamma((r-1)/2)\Gamma(r/2-1)/\Gamma(r-3/2)$, the closed form reduces to $I(r) = (r-2) 2^{r-5/2}/\sqrt{2\pi} \cdot B \cdot \Gamma(r-3/2)\Gamma(r-1/2)/\Gamma(r-1)^3$. Apply Stirling's formula to the Gamma terms and the asymptotic $B(a, b) \sim \sqrt{2\pi} a^{a-1/2} b^{b-1/2}/(a+b)^{a+b-1/2}$ when $a, b \rightarrow \infty$ with $a \sim b \sim r/2$: the Beta contributes $B((r-1)/2, r/2-1) \sim \sqrt{2\pi} 2^{-r}/\sqrt{r}$, and the Gamma-ratio $\Gamma(r-1/2)\Gamma(r-3/2)/\Gamma(r-1)^2 \sim 1$. Combining yields $I(r) \sim C/\sqrt{r}$. $\blacksquare$

Implications for the Puiseux expansion. The mean NTK expansion (12) has $t^{1/2}$ coefficient $-\frac{1}{r}[\frac{2}{\pi} + \frac{\sqrt{2}}{2\pi} I(r)]$; the $1/r$ prefactor is from the EOC recursion (Appendix B.4). The bracket $\frac{2}{\pi} + \frac{\sqrt{2}}{2\pi} I(r) \rightarrow 2/\pi$ as $r \rightarrow \infty$, recovering the shallow ReLU constant; the Fisher-Kibble correction is $O(1/(r\sqrt{r}))$. For RKHS equivalence (Bietti-Bach), only the exponent $1/2$ matters; the $1/r$ scaling does not change the RKHS.

Appendix D. Proofs for depth scaling

D.1. Correlation propagation and inverse cosine distances

For ReLU activation, the RF-LR architecture uses the same EOC parameterization as standard MLPs Hayou et al. (2018). The forward correlation map induced by ReLU is the same as in Terjék and González-Sánchez (2025); in particular, if we define a deterministic cosine recursion $\rho_k = \varrho(\rho_{k-1})$ (the infinite-width/full-rank idealization), then for ReLU $\Delta_\phi = 1/2$ (i.e. $a = b = 1/\sqrt{2}$ in (a, b) -ReLU notation). The proofs in this subsection are inspired by Terjék and González-Sánchez (2025).

Definition 31 (Cosine map for ReLU at EOC) For ReLU, set $\Delta_\phi = 1/2$. The cosine map $\varrho : [-1, 1] \rightarrow [-1, 1]$ and its derivative ϱ' satisfy

$$\varrho(\rho) = \rho + \Delta_\phi \frac{2}{\pi} (\sqrt{1-\rho^2} - \rho \arccos(\rho)), \quad \varrho'(\rho) = 1 - \Delta_\phi \frac{2}{\pi} \arccos(\rho).$$

We write $\rho_k = \varrho(\rho_{k-1})$ for the (deterministic) limiting correlation at layer k .

To a cosine $\rho \in (-1, 1)$, we associate the squared cosine distance $z = (1-\rho)/2 \in (0, 1)$ and the inverse cosine distance $w = z^{-1/2} \in (1, \infty)$.

Definition 32 (Squared and inverse cosine distance maps) Define $\zeta : [0, 1] \rightarrow [0, 1]$ by $\zeta(z) = (1-\varrho(1-2z))/2$ and $\omega : (1, \infty) \rightarrow (1, \infty)$ by $\omega(w) = \zeta(w^{-2})^{-1/2}$. Then ω is convex and satisfies

$$\omega(w) = w + \Delta_\phi \frac{4}{3\pi} + \frac{3}{2} \left(\Delta_\phi \frac{4}{3\pi} \right)^2 w^{-1} + O(w^{-2}).$$

Proposition 33 (Inverse cosine distance propagation for RF-LR) Given $w \in (1, \infty)$, for $k \in \mathbb{N}$ we have

$$|\omega^{\circ k}(w) - (w + \Delta_\phi \frac{4}{3\pi}(k-1) + \Delta_\phi \frac{2}{\pi} \log(\Delta_\phi^{-1} \frac{3\pi}{4} w + k-1))| \leq O(1).$$

Proof sketch. Step 1: Under the sequential infinite-width limit, the RF-LR correlation recursion is governed by the same ReLU cosine map ϱ as full-width MLPs (the isotropic readout factor cancels in the cosine ratio). Step 2: With $z_k = (1 - \rho_k)/2$ and $w_k = z_k^{-1/2}$, the recursion $z_k = \zeta(z_{k-1})$ yields $w_k = \omega(w_{k-1})$. Step 3: The iterate asymptotics $\omega^{\circ k}(w) \sim w + c_0(k-1) + c_1 \log(w+k)$ follow from [Terjék and González-Sánchez \(2025\)](#).

Proof This is a statement about the iterates of the inverse cosine distance map ω , so what we must justify is that ω is indeed the correct map governing the RF-LR correlation recursion.

Step 1: RF-LR correlation map (idealized) equals the ReLU cosine map. Fix a layer $\ell \geq 2$ and two inputs (x, x') . Write the RF-LR layer as

$$\mathbf{h}^{(\ell)}(x) = \frac{1}{\sqrt{n_\ell}} \sum_{j=1}^{n_\ell} A_j^{(\ell)} \boldsymbol{\sigma}(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x)), \quad \mathbf{h}^{(\ell)}(x') = \frac{1}{\sqrt{n_\ell}} \sum_{j=1}^{n_\ell} A_j^{(\ell)} \boldsymbol{\sigma}(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x')),$$

where $A_j^{(\ell)} \in \mathbb{R}^r$ is the (random) readout column and $w_j^{(\ell)}$ is the frozen random feature direction. Assume the standard RF-LR initialization: $\{A_j^{(\ell)}\}_{j=1}^{n_\ell}$ are i.i.d. centered with $\mathbb{E}[A_j^{(\ell)} A_j^{(\ell)\top}] = \sigma_A^2 I_r$, independent of $\{w_j^{(\ell)}\}_{j=1}^{n_\ell}$, and take the sequential infinite-width limit $n_\ell \rightarrow \infty$ (Definition 1).

Condition on $\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x')$. As $n_\ell \rightarrow \infty$, a conditional law of large numbers yields the usual ReLU Gaussian expectations for the (normalized) inner product and norms, and the isotropic factor $\mathbb{E}\|A_j^{(\ell)}\|^2$ cancels in the cosine ratio; thus the induced correlation recursion is the ReLU cosine map ϱ (for finite r , fluctuations occur at scale $O(r^{-1/2})$; see Appendix C.2).

Step 2: reduce to iterates of ω . With $z_k = (1 - \rho_k)/2$ and $w_k = z_k^{-1/2}$, we have $z_k = \zeta(z_{k-1})$ and therefore $w_k = \omega(w_{k-1})$, i.e. $w_k = \omega^{\circ(k-1)}(w_1)$.

Step 3: asymptotics of $\omega^{\circ k}$. The iterate asymptotic is available in ([Terjék and González-Sánchez, 2025](#), Proposition 3). ■

D.2. RF-LR NTK: recursion-driven depth scaling (not the full-MLP closed form)

For fully trained MLPs at the edge of chaos, the limiting NTK admits an explicit expression in terms of the cosine map and its derivative ([Terjék and González-Sánchez, 2025](#), Proposition 4). This relies on training all weight layers. In RF-LR, only the readouts $A^{(\ell)}$ (and biases $c^{(\ell)}$) are trained and the features $w^{(\ell)}$ are frozen; consequently the RF-LR NTK is described by our recursion (Theorem 3) and its closed form (Corollary 4, Eq. (7)) in terms of $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$.

The inverse cosine distance framework is useful here at the level of controlling the depth evolution of correlations (hence of $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$), rather than by importing the fully trained NTK identity.

D.3. Inverse cosine distance matrices and spectral bounds

To formulate dataset-level depth scaling, we work with the deterministic correlation recursion (the mean/infinite-width idealization). Given a dataset $\{x_1, \dots, x_n\} \subset \mathbb{R}^d$, set

$$\rho_{1,ij} = \frac{\langle x_i, x_j \rangle}{\|x_i\| \|x_j\|} \in [-1, 1], \quad \rho_{k,ij} = \varrho(\rho_{k-1,ij}), \quad k \geq 2.$$

Define the layer- k inverse cosine distance matrix $W_k \in \mathbb{S}^n$ by

$$(W_k)_{i,i} = 0, \quad (W_k)_{i_1,i_2} = \left(\frac{1 - \rho_{k,i_1,i_2}}{2} \right)^{-1/2}, \quad i_1 \neq i_2,$$

so that W_k is the inverse cosine distance matrix associated to ρ_k .

Proposition 34 (Spectral bounds for RF-LR inverse cosine distance matrices) *Assume EOC initialization and a dataset with no parallel pairs. For each $k \in [1 : L]$ there exists $\widehat{W}_k \in (1, \infty)$ with $\widehat{W}_k = \Theta_{W_1}(1)$ such that*

$$\|W_k - \omega^{\circ(k-1)}(\widehat{W}_k)(\mathbf{1}_n \mathbf{1}_n^\top - I_n)\| \leq O(\Delta_\phi n^{-1}k) + O(1).$$

Proof By (Terjék and González-Sánchez, 2025, Proposition 6), applied to the deterministic recursion $\rho_{k,i,j} = \varrho(\rho_{k-1,i,j})$, the inverse cosine distance matrix W_k is close to the equicorrelated structure $\omega^{\circ(k-1)}(\widehat{W}_k)(\mathbf{1}\mathbf{1}^\top - I_n)$: convexity and monotonicity of ω on $(1, \infty)$ imply the stated matrix-norm bound via standard comparison estimates for the entrywise recursion. $\blacksquare$

D.4. Preliminaries and full statements for Section 4

We work with the *deterministic proxy* recursion (84), obtained by evaluating the scalar ReLU base/derivative dual kernels along the deterministic ReLU correlation map at the edge of chaos. Starting from an initial cosine similarity $\rho_1 \in (-1, 1)$, define the associated correlation recursion by $\rho_k := \varrho^{\circ(k-1)}(\rho_1)$ for $k \geq 2$, where ϱ is the ReLU EOC cosine map. Then ρ_k aligns toward 1 at the polynomial rate $1 - \rho_k = \Theta(k^{-2})$ as in full-width MLPs at the EOC Terjék and González-Sánchez (2025); Hayou et al. (2022). The explicit proxy recursion, notation, and kernel expansions used in the proof are given below.

Scalar ReLU kernels as functions of cosine similarity. For unit-variance ReLU features, write $\theta = \arccos(\rho)$. The normalized base and derivative kernels are

$$s(\rho) = \frac{1}{2\pi}((\pi - \theta) \cos \theta + \sin \theta), \quad \dot{s}(\rho) = \frac{1}{2} - \frac{\theta}{2\pi}.$$

Near $\rho = 1$, parameterize by the inverse cosine distance $w = ((1 - \rho)/2)^{-1/2}$, so that $\rho = 1 - 2w^{-2}$ and $\theta = 2w^{-1} + O(w^{-3})$. Then

$$\dot{s}(\rho) = \frac{1}{2} - \frac{1}{\pi w} + O(w^{-3}), \quad s(\rho) = \frac{1}{2} - \frac{1}{w^2} + O(w^{-3}). \quad (83)$$

Deterministic proxy RF-LR entrywise recursion. Fix an input pair (x, x') with input cosine similarity $\rho_1 \in (-1, 1)$. The *deterministic proxy* replaces the random correlation chain $(\rho_\ell)_{\ell \geq 1}$ (see Appendix B.8) by the deterministic iterates $\rho_k := \varrho^{\circ(k-1)}(\rho_1)$ and $w_k := ((1 - \rho_k)/2)^{-1/2}$. Define the scalar deterministic proxy RF-LR NTK by

$$\Theta^{(1)}(\rho_1) = 1 + s(\rho_1), \quad \Theta^{(k)}(\rho_1) = 1 + \frac{1}{r} \Theta^{(k-1)}(\rho_1) \dot{s}(\rho_k) + \frac{1}{r} s(\rho_k), \quad k \geq 2. \quad (84)$$

This is obtained from Theorem 3 by replacing the random fields $\Sigma^{(k)}, \dot{\Sigma}^{(k)}$ with their scalar ReLU dual functions $s(\cdot), \dot{s}(\cdot)$ evaluated along the deterministic coefficients ρ_k . The proxy approximates the mean path: in the actual network, ρ_ℓ is random with $\mathbb{E}[(\rho_\ell - \varrho^{\circ(\ell-1)}(\rho_1))^2] = O(1/r)$, so for large r the random recursion concentrates around this deterministic path.

D.5. Proof of Proposition 6

Proof sketch. All entries are $\Theta(1)$. For $\lambda_{\min} \leq O(1/(rL))$: take $v = e_i - e_j$ for a pair with $\rho_{ij} < 1$; the Rayleigh quotient $v^\top \mathbf{M} v / 2 = M_{ii} + M_{jj} - 2M_{ij} = \Theta^{(L)}(1) - \Theta^{(L)}(\rho_{ij}) = \Theta(1/(rL))$ by Theorem 5. For $\lambda_{\max} = \Theta(1)$: the diagonal is $\Theta(1)$ and the matrix is positive definite. Hence $\kappa \geq \Omega(r \cdot L)$.

Proof

Entries. We have $M_{ij} = \Theta^{(L)}(\rho_{ij})$. The range is bounded and the diagonal tends to the fixed point:

$$\Theta^{(L)}(\rho) \in [\Theta^{(L)}(-1), \Theta^{(L)}(1)], \quad \Theta^{(L)}(1) \rightarrow \Theta_\star(r) = \Theta(1).$$

So all entries are $\Theta(1)$.

Upper bound on $\lambda_{\min}$: $\lambda_{\min} \leq O(1/(rL))$. By assumption (ii), there exist $i \neq j$ with $\rho_{ij} \leq \rho_{\max} < 1$. Take $v = e_i - e_j$. Then $v \perp \mathbf{1}$ and $\|v\|^2 = 2$. The Rayleigh quotient is

$$\frac{v^\top \mathbf{M} v}{\|v\|^2} = M_{ii} + M_{jj} - 2M_{ij} = \Theta^{(L)}(1) - \Theta^{(L)}(\rho_{ij}).$$

By Theorem 5, for $\rho < 1$,

$$\Theta^{(L)}(1) - \Theta^{(L)}(\rho) = \Theta(1/(rL)).$$

So this quotient is $O(1/(rL))$, hence

$$\lambda_{\min}(\mathbf{M}|_{\mathbf{1}^\perp}) \leq O(1/(rL)).$$

$\lambda_{\max} = \Theta(1)$. *Upper bound.* All eigenvalues of $\mathbf{M}|_{\mathbf{1}^\perp}$ lie in $(0, \Theta^{(L)}(1)]$ because $\mathbf{M}$ is positive definite and $M_{ii} = \Theta^{(L)}(1) = \Theta(1)$. Thus

$$\lambda_{\max} \leq \Theta(1).$$

Lower bound. Under assumption (iii) (non-euicorrelated), the off-diagonal entries $M_{ij} = \Theta^{(L)}(\rho_{ij})$ are not all equal. Pick distinct i, j, k with $\rho_{ij} \neq \rho_{ik}$. The restriction of $\mathbf{M}$ to $\text{span}\{e_i, e_j, e_k\} \cap \mathbf{1}^\perp$ is a 2×2 positive definite matrix with diagonal entries $\Theta(1)$, so its eigenvalues are $\Theta(1)$. Hence

$$\lambda_{\max}(\mathbf{M}|_{\mathbf{1}^\perp}) \geq \Omega(1).$$

Condition number.

$$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}} \geq \frac{\Theta(1)}{O(1/(rL))} = \Omega(r \cdot L).$$

■

D.6. Proof of Corollary 8

Proof

Equicorrelated data. Assume $\rho_{ij} = \rho_0$ for all $i \neq j$ and $\rho_{ii} = 1$. Then

$$\mathbf{M} = \Theta^{(L)}(1)I_n + \Theta^{(L)}(\rho_0)(\mathbf{1}\mathbf{1}^\top - I_n) = \Theta^{(L)}(\rho_0)\mathbf{1}\mathbf{1}^\top + (\Theta^{(L)}(1) - \Theta^{(L)}(\rho_0))I_n.$$

So $\mathbf{M}$ has eigenvalue $\Theta^{(L)}(1) + (n-1)\Theta^{(L)}(\rho_0)$ for the eigenvector $\mathbf{1}$, and eigenvalue $\Theta^{(L)}(1) - \Theta^{(L)}(\rho_0)$ with multiplicity $n-1$ on $\mathbf{1}^\perp$. By Theorem 5,

$$\Theta^{(L)}(1) - \Theta^{(L)}(\rho_0) = \Theta(1/(rL)).$$

On $\mathbf{1}^\perp$ all eigenvalues therefore equal $\lambda_\perp = \Theta(1/(rL))$, so

$$\kappa_\perp = 1.$$

High-dimensional random data. Let $x_1, \dots, x_n$ be i.i.d. uniform on $\mathbb{S}^{d-1}$. For $i \neq j$, $\langle x_i, x_j \rangle = \rho_{ij}$ and $\mathbb{P}(|\rho_{ij}| \geq t) \leq 2 \exp(-cdt^2)$ for $t \geq 0$ (concentration on the sphere [Vershynin \(2018\)](#)). With $\tau = C/\sqrt{d}$ for a large constant C ,

$$\max_{i \neq j} |\rho_{ij}| = O(1/\sqrt{d}) \quad \text{with probability } 1 - O(n^2) \exp(-\Omega(d)).$$

On that event, all off-diagonal ρ_{ij} lie in an interval of length $O(1/\sqrt{d})$. The map $\rho \mapsto \Theta^{(L)}(\rho)$ is Lipschitz on $[-1, 1]$, so $M_{ij} = \Theta^{(L)}(\rho_{ij})$ for $i \neq j$ differ by $O(1/\sqrt{d})$. Thus $\mathbf{M}$ equals an equicorrelated matrix (common off-diagonal $\Theta^{(L)}(\bar{\rho})$ for some $\bar{\rho} = O(1/\sqrt{d})$) plus E with

$$\|E\|_F \leq n^2 \cdot O(1/\sqrt{d}) = o(1) \quad \text{as } d \rightarrow \infty \text{ with } n \text{ fixed.}$$

By Weyl's inequality, each eigenvalue of $\mathbf{M}|_{\mathbf{1}^\perp}$ differs from

$$\lambda_\perp^0 = \Theta^{(L)}(1) - \Theta^{(L)}(\bar{\rho})$$

by $O(\|E\|_{\text{op}}) = o(1)$. For $\bar{\rho} = O(1/\sqrt{d})$, the proxy recursion gives $\Theta^{(L)}(\bar{\rho})$ bounded and $\Theta^{(L)}(1) = \Theta_\star(r) + O(1/L) = \Theta(1)$, so $\lambda_\perp^0$ is $\Theta(1)$ (for fixed r, L ; the r, L -scaling is $\Theta(1/(rL))$ by the depth-induced gap). Hence

$$\text{each eigenvalue of } \mathbf{M}|_{\mathbf{1}^\perp} = \lambda_\perp^0 + o(1) = \Theta(1)(1 + o(1)), \quad \kappa_\perp = 1 + o(1).$$

On that high-probability event the data are approximately equicorrelated, so the same operator-norm bound as in Theorem 7 applies to $\hat{K}$:

$$\|(\hat{K} - K_{\text{proxy}})|_{\mathbf{1}^\perp}\|_{\text{op}} = O_P(L/r + 1/\sqrt{r}) \quad \text{with high probability (in } r \text{ and } d).$$

■

Terjék–González–Sánchez pipeline and RF-LR. Terjék and González–Sánchez (2025) obtain sharp two-sided eigenvalue and condition-number bounds for full-width MLP NTKs via a three-step pipeline:

- (i) Approximate NTK entries by an affine function of the inverse cosine distance $\omega^{\circ(l-1)}(w)$;
- (ii) Show the inverse cosine distance matrix is close to an equicorrelated structure;
- (iii) Transfer spectral bounds via Weyl.

For RF-LR we have (ii) in Proposition 34, but (i) fails: the RF-LR scalar recursion

$$\Theta^{(k)}(\rho) = 1 + \frac{1}{r} \Theta^{(k-1)} \dot{s}(\rho_k) + \frac{1}{r} s(\rho_k)$$

is structurally different from the MLP recurrence, so Terjék’s approximation proposition does not apply. We do not have an RF-LR result of the form

$$\Theta^{(L)}(\rho) \approx A(r) + B(r) \cdot \omega^{\circ(L-1)}(((1-\rho)/2)^{-1/2})/r$$

with explicit $A(r), B(r)$ and $O(1/(rL^2))$ error. Given such an approximation, (iii) would mirror Terjék: the Gram matrix would approximate a rank-one perturbation of a diagonal matrix, yielding sharp $\lambda_{\min}, \lambda_{\max}$ and κ bounds.

Equicorrelated vs general datasets. For *equicorrelated* datasets ($\rho_{ij} = \rho_0$ for $i \neq j$), the mean kernel matrix

$$K = \Theta^{(L)}(1)I + \Theta^{(L)}(\rho_0)(\mathbf{1}\mathbf{1}^\top - I)$$

has explicit spectrum on $\mathbf{1}^\perp$, and sharp bounds hold with $\Theta^{(L)}(1) - \Theta^{(L)}(\rho_0) \asymp 1/(rL)$. For general datasets we are limited to one-sided bounds:

$$\lambda_{\min} \leq O(1/(rL)), \quad \lambda_{\max} = \Theta(1), \quad \kappa \geq \Omega(rL).$$

A lower bound on $\lambda_{\min}$ and an upper bound on κ would require either an RF-LR analogue of Terjék’s NTK–inverse-cosine approximation (e.g. by analyzing the recursion with $\dot{s}(\rho_k) \approx 1/2 - 1/(\pi w_k)$, $s(\rho_k) \approx 1/2 - 1/w_k^2$, and matching asymptotic style) or a perturbation argument for datasets with mild variation in ρ_{ij} .

D.7. Proof of Theorem 5

Proof sketch. Step 1 (correlation alignment): Proposition 33 gives $w_k = \omega^{\circ(k-1)}(w_1) \sim c_- k$, hence $1 - \rho_k = 2w_k^{-2} = \Theta(k^{-2})$. Step 2 (kernel saturation): Rewrite the recursion as $\Theta_k = b_k + a_k \Theta_{k-1}$; the limiting coefficients $a = 1/(2r)$, $b = 1 + 1/(2r)$ yield fixed point $\Theta_\star(r) = (2r + 1)/(2r - 1)$. A discrete Grönwall argument shows $|\Theta_k - \Theta_\star| = O(1/k)$. Step 3 (gap decay): Subtract the diagonal and off-diagonal recursions; the gap $\Delta_k = \Theta_k^{\text{diag}} - \Theta_k^{\text{off}}$ satisfies $\Delta_k = a\Delta_{k-1} + \Theta(1/(rk))$, hence $\Delta_k = \Theta(1/(rk))$.

Proof

Step 1: correlation alignment. By Proposition 33, there exists a constant $C_w < \infty$ such that for all $k \geq 1$,

$$w_k = \omega^{\circ(k-1)}(w_1) = w_1 + c_0(k-1) + c_1 \log(c_2 w_1 + k - 1) + \varepsilon_k, \quad |\varepsilon_k| \leq C_w,$$

where $c_0 = \Delta_\phi \frac{4}{3\pi} > 0$, $c_1 = \Delta_\phi \frac{2}{\pi}$, $c_2 = \Delta_\phi^{-1} \frac{3\pi}{4}$.

In particular, $w_k \geq w_1 + c_0(k-1) - C_w$. Choosing k_0 large enough gives $w_k \geq \frac{c_0}{2}k$ for all $k \geq k_0$. Similarly, $w_k \leq c_+k$ for some $c_+ < \infty$ and all $k \geq k_0$. Thus $c_-k \leq w_k \leq c_+k$ for $k \geq k_0$, and $1 - \rho_k = 2w_k^{-2} = O(k^{-2})$.

Step 2: kernel saturation. Write $\Theta_k = \Theta^{(k)}(\rho_1)$ and rewrite (84) as

$$\Theta_k = b_k + a_k \Theta_{k-1}, \quad a_k = \frac{1}{r} \dot{s}(\rho_k), \quad b_k = 1 + \frac{1}{r} s(\rho_k).$$

For the limiting constant-coefficient recursion (corresponding to $\rho_k \rightarrow 1$), we have $a = \frac{1}{r} \dot{s}(1) = \frac{1}{2r}$, $b = 1 + \frac{1}{r} s(1) = 1 + \frac{1}{2r}$, hence $\Theta_*(r) = b/(1-a) = (2r+1)/(2r-1)$.

Set $e_k = \Theta_k - \Theta_*(r)$. Then

$$e_k = a_k e_{k-1} + (b_k - b) + (a_k - a) \Theta_*(r).$$

Using (83) and $w_k \geq c_-k$ for $k \geq k_0$, there exists $C < \infty$ such that for all $k \geq k_0$,

$$\begin{aligned} |a_k - a| &= \frac{1}{r} \left| \dot{s}(\rho_k) - \frac{1}{2} \right| \leq \frac{C}{r} \frac{1}{w_k} \leq \frac{C}{r} \frac{1}{k}, \\ |b_k - b| &= \frac{1}{r} \left| s(\rho_k) - \frac{1}{2} \right| \leq \frac{C}{r} \frac{1}{w_k^2} \leq \frac{C}{r} \frac{1}{k^2}. \end{aligned}$$

Moreover $0 \leq a_k \leq a = 1/(2r) < 1$. Hence for all $k \geq k_0$,

$$|e_k| \leq a |e_{k-1}| + \frac{C'}{k}$$

for some $C' < \infty$. A standard induction (discrete Grönwall) yields $|e_k| \leq C_\Theta/k$ for all $k \geq 1$; in particular $e_k \rightarrow 0$ and $\Theta_k \rightarrow \Theta_*(r)$.

To make the $O(1/k)$ rate explicit, fix $k \geq k_0$ and iterate the one-step inequality:

$$|e_k| \leq a^{k-k_0} |e_{k_0}| + \sum_{j=k_0+1}^k a^{k-j} \frac{C'}{j}.$$

Since $j \geq k_0 + 1$ implies $1/j \leq 1/k_0$ and also $j \leq k$ implies $1/j \leq 1/k$, we can bound

$$\sum_{j=k_0+1}^k a^{k-j} \frac{1}{j} \leq \frac{1}{k} \sum_{t=0}^{k-k_0-1} a^t \leq \frac{1}{k} \cdot \frac{1}{1-a}.$$

Also $a^{k-k_0} |e_{k_0}| \leq |e_{k_0}| \leq (k_0 |e_{k_0}|)/k$. Therefore for all $k \geq k_0$,

$$|e_k| \leq \frac{1}{k} \left(k_0 |e_{k_0}| + \frac{C'}{1-a} \right).$$

Absorbing the finitely many values $k < k_0$ into the constant gives $|e_k| \leq C_\Theta/k$ for all $k \geq 1$, as claimed.

Step 3: diagonal–off-diagonal gap. Let $\Theta_k^{\text{diag}} = \Theta^{(k)}(1)$ and $\Theta_k^{\text{off}} = \Theta^{(k)}(\rho_1)$, and set $\Delta_k = \Theta_k^{\text{diag}} - \Theta_k^{\text{off}} \geq 0$. The diagonal recursion is $\Theta_k^{\text{diag}} = b + a \Theta_{k-1}^{\text{diag}}$, while the off-diagonal recursion is $\Theta_k^{\text{off}} = b_k + a_k \Theta_{k-1}^{\text{off}}$. Subtracting gives the exact identity

$$\Delta_k := a \Delta_{k-1} + (a - a_k) \Theta_{k-1}^{\text{off}} + (b - b_k).$$

For k large, $\Theta_{k-1}^{\text{off}}$ is bounded above and below by positive constants (since $\Theta_k^{\text{off}} \rightarrow \Theta_\star(r) \in (1, \infty)$). Also by (83) and $w_k \asymp k$,

$$a - a_k = \frac{1}{r} \left(\frac{1}{2} - \dot{s}(\rho_k) \right) = \Theta \left(\frac{1}{r w_k} \right) = \Theta \left(\frac{1}{r k} \right), \quad b - b_k = \frac{1}{r} \left(\frac{1}{2} - s(\rho_k) \right) = O \left(\frac{1}{r w_k^2} \right) = O \left(\frac{1}{r k^2} \right).$$

Thus there exist k_1 and constants $0 < c < C < \infty$ such that for all $k \geq k_1$,

$$a \Delta_{k-1} + \frac{c}{r k} \leq \Delta_k \leq a \Delta_{k-1} + \frac{C}{r k}.$$

Iterating these comparison recursions yields $c_\Delta/(rk) \leq \Delta_k \leq C_\Delta/(rk)$ for all $k \geq k_1$, completing the proof. $\blacksquare$

D.8. Remark: proxy lower bounds vs positivity (smallest eigenvalue)

Notation. Let $\widehat{K}$ denote the empirical NTK Gram matrix at initialization, with entries $\widehat{K}_{ij} = \langle \nabla_\theta f(x_i), \nabla_\theta f(x_j) \rangle$. Let K denote the mean (deterministic) kernel Gram matrix with entries $K_{ij} = \mathbb{E}[\widehat{K}_{ij}]$. The *centered* spectrum refers to the restriction of these matrices to the subspace $\mathbf{1}^\perp$ of vectors orthogonal to the constant vector $\mathbf{1}$; we write $\widehat{K}|_{\mathbf{1}^\perp}$ and $K|_{\mathbf{1}^\perp}$ for these restrictions.

Remark 35 (Can the smallest eigenvalue become negative?) *No: $\widehat{K}$ is a Gram matrix of gradients, hence $\widehat{K} \succeq 0$ deterministically and all its eigenvalues are ≥ 0 . Restricting to the mean-zero subspace preserves positive semidefiniteness: for any $v \perp \mathbf{1}$, one has $v^\top \widehat{K} v \geq 0$, so $\widehat{K}|_{\mathbf{1}^\perp} \succeq 0$.*

What can happen is that a deterministic proxy lower bound on the centered eigenvalue scale is smaller than the magnitude of random fluctuations. Weyl’s inequality gives

$$\lambda_{\min}(\widehat{K}|_{\mathbf{1}^\perp}) \geq \lambda_{\min}(K|_{\mathbf{1}^\perp}) - \|\widehat{K} - K\|_2,$$

so if $\|\widehat{K} - K\|_2$ dominates the deterministic scale of $\lambda_{\min}(K|_{\mathbf{1}^\perp})$, the bound becomes vacuous (it may be negative), but the true eigenvalue still satisfies $\lambda_{\min}(\widehat{K}|_{\mathbf{1}^\perp}) \geq 0$. Thus, the relevant concern is not “going below zero” but rather “being driven close to zero” when the $1/(rL)$ signal is swamped by noise.

D.9. Remark: infinite-depth limit after centering

Remark 36 (Infinite-depth limit: existence and degeneracy after centering) *The correlation recursion satisfies $1 - \rho_k = O(k^{-2})$ as in full-width MLPs at the EOC [Terjék and González-Sánchez \(2025\)](#); [Hayou et al. \(2022\)](#), so deep features become increasingly aligned. For the mean RF-LR NTK recursion (84), Theorem 5 implies that for any fixed $\rho_1 \in (-1, 1)$,*

$$\Theta^{(L)}(\rho_1) \rightarrow \Theta_\star(r) \quad \text{and} \quad \Theta^{(L)}(1) \rightarrow \Theta_\star(r),$$

so the entrywise infinite-depth limit of the (mean) Gram matrix is a constant kernel: $K^{(L)} \rightarrow \Theta_\star(r) \mathbf{1}\mathbf{1}^\top$. Consequently, after centering (restricting to the orthogonal complement of the constant mode), the relevant part of $K^{(L)}$ tends to zero and the spectrum collapses. In particular, a non-degenerate deep limit on $\mathbf{1}^\perp$ can only hold under a joint scaling (e.g. r growing with L) or after an explicit rescaling of the centered operator (equivalently, rescaling time/learning rate in kernel gradient descent).

Appendix E. Experiments

This section presents numerical illustrations. All scripts are headless (matplotlib “Agg” backend). Setup: finite-width NTK scripts use a **3-layer (2 ReLU)** RF-LR architecture with hidden width 12000–20000 so that finite-width variance is small; deterministic proxy scripts iterate the recursion to large k (e.g. $k_{\max} = 4000$ or $L = 200$). Bottleneck rank r is swept as indicated; n is the number of inputs, d the input dimension. Scripts in the first subsection are **deterministic**; the rest are **finite-width Monte Carlo** (they illustrate fluctuation scales and do not constitute proofs).

E.1. Correlation alignment and equicorrelated spectrum (Theorem 5, Corollary 8)

What is computed. Deterministic ReLU EOC correlation recursion $\rho_k = \varrho(\rho_{k-1})$ and proxy NTK recursion; diagnostics w_k/k , $k|\Theta^{(k)} - \Theta_\star(r)|$, $rk(\Theta_{\text{diag}}^{(k)} - \Theta_{\text{off}}^{(k)})$. For the equicorrelated model, eigenvalues $\lambda_1 = \Theta^{(L)}(1) + (n-1)\Theta^{(L)}(\rho_0)$, $\lambda_\perp = \Theta^{(L)}(1) - \Theta^{(L)}(\rho_0)$ (e.g. $n = 64$, $\rho_0 = 0$).

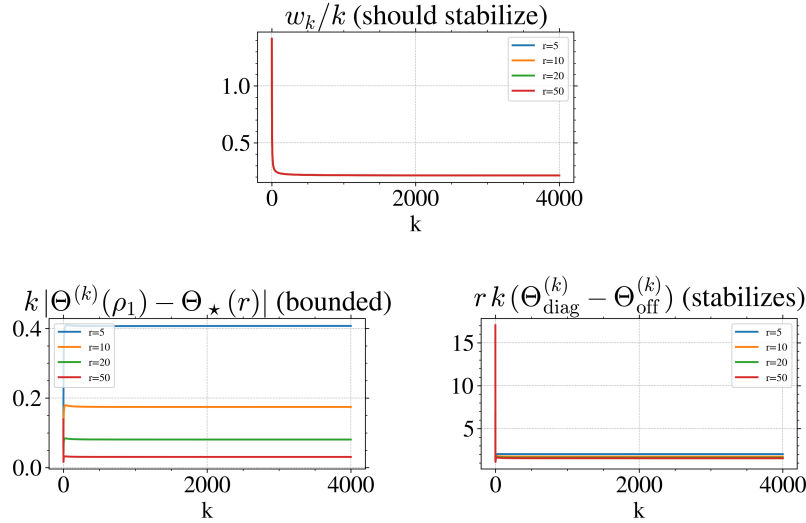


Figure 2: Three panels (Theorem 5): w_k/k , $k|\Theta^{(k)} - \Theta_\star(r)|$, $rk(\Theta_{\text{diag}}^{(k)} - \Theta_{\text{off}}^{(k)})$ vs k . One curve per $r \in \{5, 10, 20, 50\}$.

E.2. Product decay and entry variance

What is computed. Left: Deterministic product $\prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}/r$ vs $j = L - \ell$ (exponential-in-depth suppression, Eq. 7). **Right:** Monte Carlo $\text{Var}[K(-1, 1)]$ vs r over 100 initializations (3-layer RF-LR, width 16000); rank-driven concentration (Section 5).

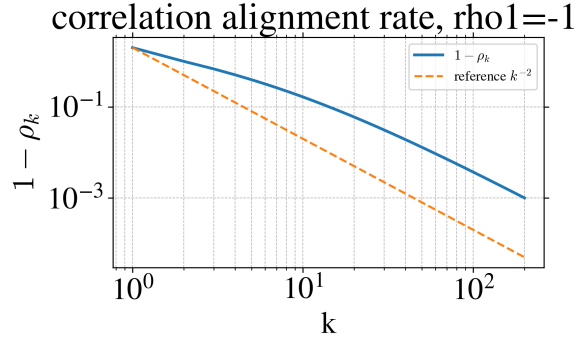


Figure 3: Correlation alignment along depth: $1 - \rho_k = O(k^{-2})$ (log-log) as in Theorem 5.

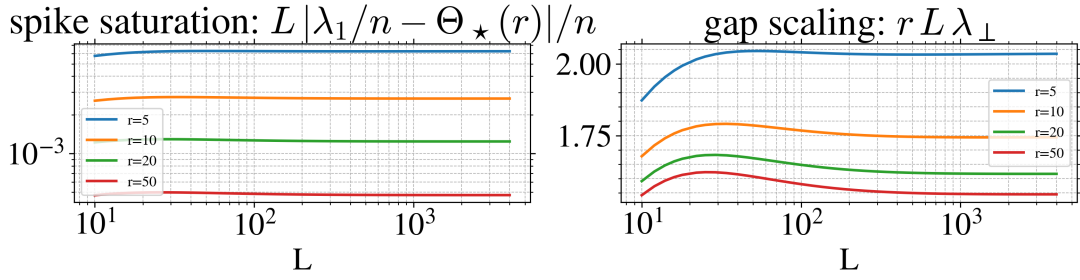


Figure 4: Equicorrelated spectrum (Corollary 8): (1) spike saturation $L|\lambda_1/n - \Theta_\star(r)|/n$ vs L (log-log); (2) gap scaling $rL\lambda_\perp$ vs L (log-log). One curve per r .

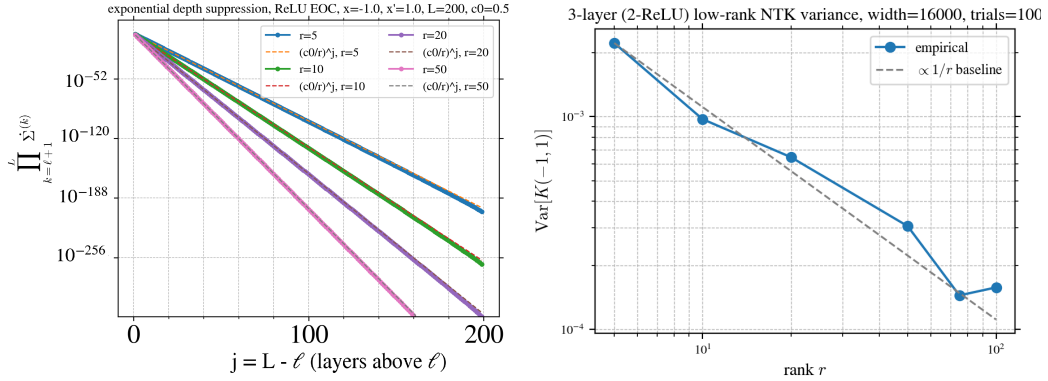


Figure 5: **Left:** Product decay vs depth; bound $(c_0/r)^j$. **Right:** Entry variance vs r ; $\propto 1/r$ baseline.

E.3. Smallest eigenvalue (finite-width Monte Carlo)

What is computed. Fixed random dataset on the unit sphere ($n = 64, d = 16$); empirical Gram K , centered $K_c = HKH$; $\lambda_{\min}^+(K_c)$ over 50 initializations, $r \in \{5, 10, 20, 50, 100, 200, 500, 1000\}$. Illustrates smallest eigenvalue driven close to 0 by finite-width noise (Appendix 35).

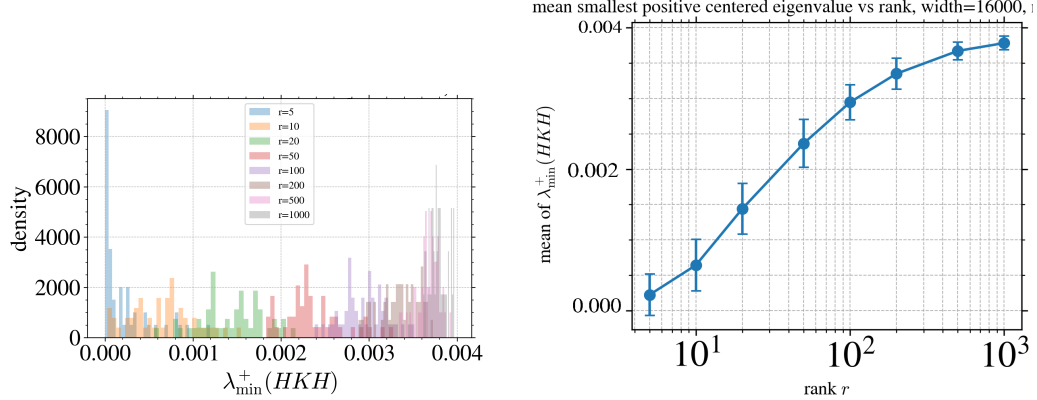


Figure 6: **Left:** Histograms of $\lambda_{\min}^+(HKH)$; matrix stays PSD. **Right:** Mean $\lambda_{\min}^+(HKH)$ vs r ; better conditioning with r .

E.4. Condition number κ (proxy vs empirical; equicorrelated, high-dim, non-equicorrelated)

What is computed. Condition number on $1^\perp$ vs r . **Equicorrelated:** $n = 32, d = 64, \rho_0 = 0$; proxy $\kappa_\perp = 1$ (Corollary 8); empirical mean $\pm$ std over 40 initializations (Theorem 7). **High-dim spherical:** i.i.d. uniform on S^{d-1} , $d = 256, 128$; $\kappa_\perp \approx 1$. **Non-equicorrelated:** Clustered design (4 clusters), $n = 48, d = 64, 30$ trials; $\kappa \geq \Omega(r \cdot L)$ (Proposition 6).

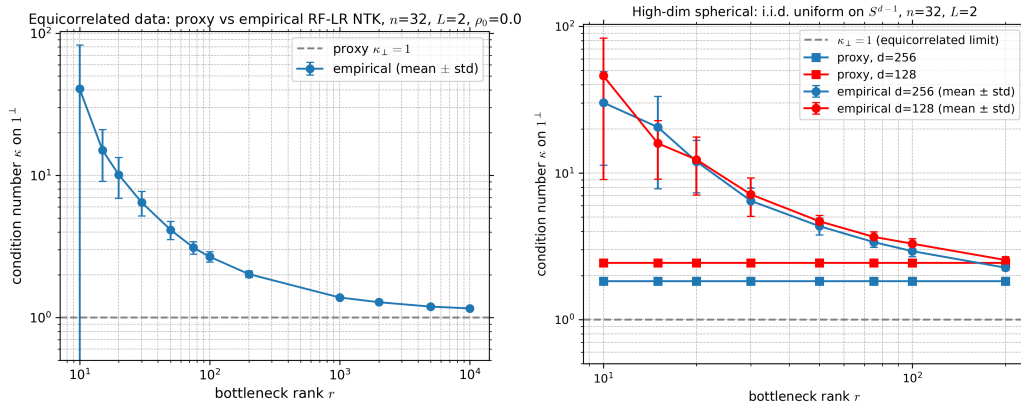


Figure 7: **Left:** Equicorrelated: proxy $\kappa_\perp = 1$; empirical concentrates. **Right:** High-dim spherical: $\kappa_\perp \approx 1$.

E.5. Non-equicorrelated κ and kernel regression risk

Left: Condition number vs r for clustered (non-equicorrelated) data; κ need not approach 1 (Proposition 6). **Right:** Kernel ridge regression with the empirical RF-LR NTK: fix a target on the sphere, 64 train / 256 test, $d = 32$, 20 trials; as r grows the Gram concentrates toward the proxy, so test MSE and its variance across trials decrease, linking conditioning and concentration to downstream risk.

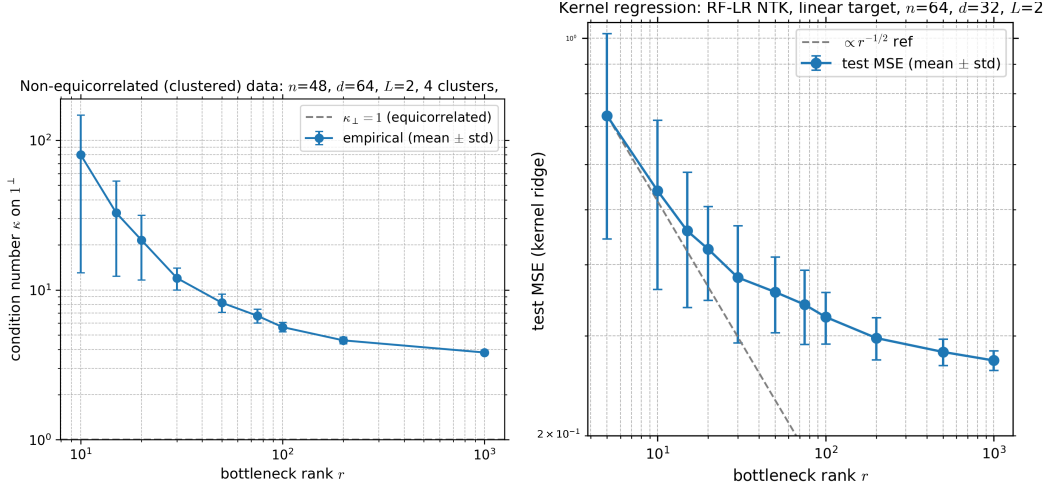


Figure 8: **Left:** κ vs r , non-equicorrelated; $\kappa \geq \Omega(r \cdot L)$. **Right:** Test MSE vs r ; mean risk and std drop with r .

E.6. RKHS Puiseux exponent vs depth (Corollary 13, extension to $L \geq 4$)

For zonal kernels on the sphere, the RKHS is controlled by the Puiseux exponent γ at the endpoint $\rho = 1$: if $K(1-t) - K(1) \sim c \cdot t^\gamma$ for small t , then γ determines the RKHS [Bietti and Bach \(2021\)](#). The paper proves $\gamma = 1/2$ for the mean three-layer RF-LR kernel (same as shallow ReLU); extension to $L \geq 4$ is open. We estimate $\gamma(L)$ for the deterministic proxy $\Theta^{(L)}(\rho)$ via log-log regression of the gap $\Theta^{(L)}(1) - \Theta^{(L)}(1-t)$ vs t . For $L \in \{2, 3, 4, 5, 6\}$ we obtain $\gamma(L) \approx 0.52$ – 0.55 with $R^2 > 0.999$, suggesting the proxy’s endpoint behavior is consistent with RKHS equivalence extending to $L \geq 4$.

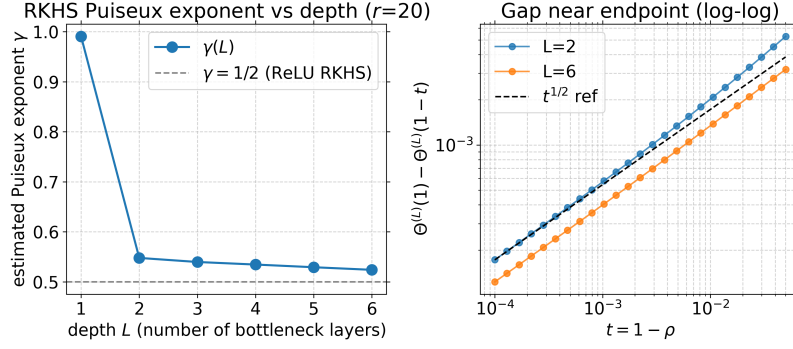


Figure 9: RKHS Puiseux exponent vs depth. Left: estimated $\gamma(L)$ vs L ; reference $\gamma = 1/2$ (ReLU RKHS). Right: log-log gap vs t for $L = 2$ and $L = 6$. The proxy kernel exhibits $\gamma \approx 1/2$ for $L \geq 2$, consistent with RKHS equivalence extending beyond three layers.